

FINITE CERTIFICATES, DIAGRAMMATIC GLOBALIZATION, AND TYPED RETURNS THE MATHEMATICAL KERNEL OF AK HIGH-DIMENSIONAL PROJECTION STRUCTURAL THEORY

ATSUSHI KOBAYASHI

ABSTRACT. We formulate *high-dimensional projection structural analysis* as a typed mathematical passage from a source object X to a structured shadow $F_X = \mathfrak{R}_{\text{AK}}(X)$, from that shadow to finite collapse certificates and reconstructed diagrams, and from the resulting internal objects back to a source-domain conclusion by a separately proved Return theorem. The word projection is structural: it may be objectwise or partially functorial, and no global faithfulness, exactness, or literal reduction of topological dimension is assumed. Instead, the projection is required to be purpose-faithful for a declared feature, direction, source class, and strength. This makes the central asymmetry explicit: projection exposes a controlled shadow, whereas Return alone carries a certified statement back to the source.

The mathematical kernel has two interacting but non-identical cores. The persistence core treats constructible one-parameter shadows, repaired hard-threshold evaluation, finite detector certificates, finite-to-infinite reduction, terminal and transient collapse defects, and model-relative certificate complexity. The diagrammatic core treats actual finite diagrams, natural transformations, projective generators, cones, finite Morita reconstruction, finite Alexandrov realization, and effective descent. A typed interface separates robust but lossy observation from exact categorical reconstruction.

Several results are mathematically sharp within their declared models. Hard threshold deletion satisfies a two-sided threshold-gap stability theorem and exact zero separation on the reduced image, while global non-expansiveness fails. When finite family-relative schemes exist, certificate complexity is represented by a finite Pareto frontier in the four coordinates of query count, bit length, verification work, and artifact size; private-witness families give exact information-model lower bounds, and a right-unbounded birth family admits no finite exact detector. Endpoint-complete persistence is reconstructed by finite Möbius inversion, and the full rank table is query-minimal in the coordinate rank-query model. Typed finite-state inverse towers satisfy

$$\varprojlim V_n \cong \text{im } T^e, \quad \varprojlim^1 V_n = 0,$$

with an exact finite apex-agreement test. Actual finite diagrams admit generator-based zero reflection, cone detection, Morita reconstruction, and strict finite descent. Finally, a source-bound proof DAG returns only the target-relative strength supported by every required component. When a finite target slice contains required nodes of atomic status *reject* or *undefined*, their earliest causal failures form a unique finite antichain; a certified nonzero obstruction is instead a mathematical output and is not relabelled as missing evidence.

This article is a standalone mathematical projection of AK-HPDST v21.0.0. It claims neither unrestricted multiparameter classification, arbitrary higher-stack descent, unbounded-derived reconstruction, nor a solution of any named external open problem.

Date: August 2026.

2020 Mathematics Subject Classification. 55N31, 18E10, 18G80, 16G20, 18F20, 68Q25.

Key words and phrases. high-dimensional projection structural analysis, persistent homology, persistence modules, finite proof certificates, Collapse diagnostics, finite diagrams, projective generators, Morita reconstruction, Alexandrov spaces, descent, typed returns.

CONTENTS

1. Introduction and main results	4
1.1. The projection problem behind finite proof	4
1.2. The dual-core architecture	4
1.3. The projection principle and four synthesis theorems	5
1.4. Contribution and relation to standard ingredients	6
1.5. Scope and source convention	6
2. High-dimensional projection structural analysis	6
2.1. Purpose-faithful structural projection	6
2.2. Structured shadows and loss profiles	7
2.3. Collapse as a typed structural analysis	8
2.4. Projection–Return asymmetry	8
2.5. Mathematically distinguished results	9
3. Relation to existing literature and mathematical positioning	11
3.1. Persistence modules, barcodes, and stability	11
3.2. Finite quivers, projective probes, and Morita reconstruction	11
3.3. Finite spaces, sheaves, and descent	11
3.4. Position of the integrated framework	12
4. Standing notation and dependency convention	12
5. Typed architecture and proof semantics	13
5.1. Persistence objects and finite diagrams	13
5.2. The persistence-to-diagram interface	13
5.3. Record globalization and object globalization	14
5.4. Proof DAGs and non-compensation	14
5.5. Statuses and result sorts	14
6. Repaired persistence and finite certificates	15
6.1. Barcode conventions and hard threshold deletion	15
6.2. Failure of global non-expansiveness	16
6.3. Sharp threshold-gap stability	16
6.4. Zero separation on the reduced image	17
6.5. Window-first evaluation	18
6.6. Certificate models	18
6.7. A common-model exact benchmark	19
6.8. Right-unbounded no-go theorem	19
6.9. Composition and evidence reuse	20
7. Finite-to-infinite reduction and record-level globalization	20
7.1. Tower variance and the finite-prefix boundary	20
7.2. Finite-state inverse limits	21
7.3. Finite-to-infinite soundness and independent defects	22
7.4. Certificate atlases and global record zero	23
8. Finite diagram reconstruction	23
8.1. Endpoint-complete atom grids	23
8.2. Endpoint-complete reconstruction	24
8.3. Rank tables and finite Möbius inversion	25
8.4. Enriched probes and morphism reconstruction	26
8.5. P2DG handoff	26
8.6. Exact coordinate-query complexity	27
9. Finite diagrammatic globalization	28

9.1. Finite category presentations	28
9.2. The abelian finite-diagram category	28
9.3. Representable projectives and the finite generator	29
9.4. Derived zero reflection and cone detection	29
9.5. Finite Morita reconstruction	30
9.6. Finite Alexandrov realization	31
9.7. Effective strict descent	31
9.8. Bounded-complex descent	32
9.9. Global zero and isomorphism certificates	32
10. Typed Bridge composition and Return	33
10.1. Universal Bridge data	33
10.2. Assumption and strength propagation	34
10.3. Partial and regularized Returns	35
10.4. Rigidity and promotion	35
10.5. Infinite quantifiers and tail selection	35
10.6. The unified defect stack	36
10.7. Unified closure	36
10.8. Alternative routes and failure isolation	37
11. Kernel demonstrations and exact complexity	38
11.1. A complete rank certificate on $A_{[3]}$	38
11.2. Why scalar probes are not enriched reconstruction	39
11.3. Object and bounded-complex descent on a finite poset	40
11.4. A complete Return through both cores	40
11.5. Return ceiling	41
11.6. Validation by paired positive and negative cases	41
12. Scope, contribution, and research program	42
12.1. What the kernel proves	42
12.2. What is outside the stable scope	42
12.3. Where the integrated contribution lies	43
12.4. Next steps	43
Appendix A. Canonical source and theorem map	44
Appendix B. Counterexample suite and proof boundaries	45
B.1. Threshold crossing	45
B.2. Incomplete finite observation	45
B.3. Omitted relations	45
B.4. Scalar probes	45
B.5. Objectwise isomorphism	45
B.6. Pairwise overlap maps	46
B.7. Effectivity without conservativity	46
B.8. Predicate Return without source reconstruction	46
B.9. Finite-prefix and infinite-target separation	46
Appendix C. Compact record schemas	46
C.1. Certificate record	46
C.2. P2DG record	46
C.3. Descent record	46
C.4. Bridge record	46
Data and source availability	46
Versioning note	47
References	47

1. INTRODUCTION AND MAIN RESULTS

1.1. The projection problem behind finite proof. The motivating objects of AK-HPDST need not be persistence modules. They may be filtered complexes, arithmetic towers, analytic solution families, derived objects, categorical systems, or geometric degenerations whose native structure is too large, too high-dimensional, or too nonlocal for direct finite verification. The first step is therefore not a barcode computation but a declared structural projection

$$X \rightsquigarrow F_X = \mathfrak{R}_{\text{AK}}(X)$$

into a Core-readable object. The wavy arrow is deliberate: the construction may initially be objectwise or partially functorial. It may be written as a genuine functor only after variance and composition have been proved.

A projection is useful only relative to a purpose. One must state which source feature is represented by which feature of F_X , in which direction the implication is used, on what source class, under what coefficient and parameter conventions, and with what failure boundary. Without that purpose-faithfulness theorem, the projected object is a candidate shadow rather than theorem-bearing evidence.

After projection, a large class of arguments in topology, representation theory, homological algebra, arithmetic, and analysis has the same vulnerable shape. One computes finitely many values, maps, ranks, local objects, or overlap comparisons; recognizes a stable pattern; and promotes the pattern to a global or source-domain conclusion. The vulnerable step is not usually the finite computation. It is the promotion from observation to proof, from a shadow to an actual diagram, from local data to a global object, and from an internal conclusion back to the source. Standard persistence, quiver, and homological-algebraic ingredients used below are reviewed in [section 3](#); see also [\[6, 5, 12, 7, 13\]](#).

The complete kernel chain is

$$\begin{aligned} \text{high-dimensional source } X &\xrightarrow{\text{typed projection}} F_X = \mathfrak{R}_{\text{AK}}(X) \\ &\xrightarrow{\text{persistent shadow}} B_{n,W}(F_X) \xrightarrow{\text{repaired collapse}} \mathbf{C}_{\text{fin}}(X) \\ &\xrightarrow{\text{P2DG}} \text{actual finite diagram} \xrightarrow{\text{effective descent}} \text{global object} \\ &\xrightarrow{\text{typed Return}} E_\sigma(X). \end{aligned} \tag{1}$$

Every mathematical arrow in (1) has an independent source type, target type, direction, hypothesis set, conclusion strength, and failure mode. The source-controlled v21 route also contains a distinct identity/audit edge, discussed in [Theorems 2.9](#) and [10.12](#); that edge governs admissible invocation rather than mathematical strength. No arrow is licensed by terminology alone. In particular, a finite list of observations is not a proof certificate merely because it is short or reproducible; a barcode or rank table is not an actual diagram with morphisms; pairwise overlap equivalences are not a cocycle; and an internal collapse statement is not an external theorem without a Return map.

1.2. The dual-core architecture. The stable mathematical baseline is

$$\text{Core}_P + \text{Core}_{DG}.$$

The symbol $+$ denotes coordinated coexistence through typed interfaces. It does not denote a product, coproduct, equivalence, localization, or automatic mutual reconstruction.

The persistence core Core_P is built on constructible one-parameter persistence over a field. It contains window-first evaluation, hard threshold deletion, sharp threshold-gap transport,

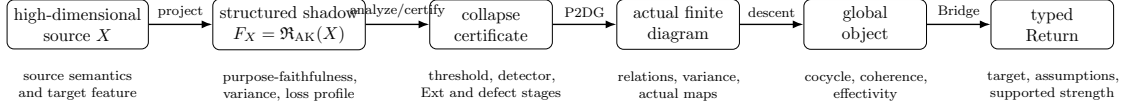


FIGURE 1. High-dimensional projection structural analysis. Projection produces a purpose-relative structured shadow; only separately proved reconstruction, descent, rigidity, and Return arrows can carry a conclusion back to the source domain.

finite detector certification, model-relative certificate complexity, and certified finite-to-infinite reductions. The diagrammatic core Core_{DG} is built on finite categories or finite posets, finite-dimensional diagrams, bounded complexes, actual natural transformations, projective generators, cones, finite Morita reconstruction, Alexandrov models, and explicitly effective finite descent.

The interface between the two cores records more than a map on objects. It records the source and target categories, variance, action on objects and morphisms, exactness, reflection, information preservation, terminal conventions, and the strongest admissible downstream use. This is essential because object reconstruction, morphism reconstruction, exact reconstruction, and derived reconstruction are distinct strengths.

1.3. The projection principle and four synthesis theorems. The paper is organized around one governing projection-and-return principle and four source-bound synthesis theorems. They repackage canonical results of AK-HPDST v21.0.0 into a standalone article; they do not create a second theorem authority.

Principle P: high-dimensional projection and asymmetric Return. A theorem-bearing AK route starts from a source object X , constructs a purpose-faithful shadow $F_X = \mathfrak{R}_{AK}(X)$, certifies the relevant collapse or obstruction statement inside the dual Core, and returns a source-domain conclusion only through a separate theorem at a declared strength. The projection need not be globally faithful. Its precise preservation and reflection burden is part of the theorem, not part of the notation.

Theorem A: repaired finite certificates. Hard positive-threshold deletion is neither a global exact localization nor globally non-expansive for the bottleneck metric. It becomes stable under a sharp two-sided exclusion band around the threshold. On the reduced image, the same bound separates zero from nonzero profiles. Together with an explicit certificate model this yields exact finite query bounds on bounded promised families and an impossibility theorem on right-unbounded birth families.

Theorem B: finite diagram reconstruction and complexity. An endpoint-complete finite grid turns a constructible one-parameter persistence module into a finite linear-quiver representation without loss on the declared class. A complete rank table recovers interval multiplicities by a finite Möbius formula. In the coordinate rank-query model, the number of comparable pairs is necessary and sufficient for exact reconstruction of the unrestricted interval-multiplicity family.

Theorem C: finite diagrammatic globalization. For a finite category P , the sum of the linearized covariant representables is a projective generator of $\text{Diag}_P(\mathbb{k})$. Its derived Hom functor reflects zero, and its application to a cone detects isomorphisms. The enriched probe yields an exact Morita reconstruction. For finite posets, diagrams are realized as sheaves on the upper Alexandrov space; actual cocycle data then give strict finite object and bounded-complex descent.

Theorem D: typed end-to-end Return. A mathematical Bridge DAG closes only when every required realization, coefficient, core, reduction, descent, coherence, rigidity, and final

interpretation node is available at the consumed type. The resulting Return is valid only at the exact target record, under the propagated assumption context, and at a strength contained in every necessary component's supported lower set. The source-controlled v21 label *AK theorem* additionally requires source binding, quarantine, identity, and audit closure; those governance predicates license the recorded use of the theorem but do not strengthen its mathematical conclusion. Missing mathematical arrows cannot be compensated by metric reserve, search coverage, replay success, or agreement among agents.

1.4. Contribution and relation to standard ingredients. The individual ingredients include standard persistence decomposition and stability, representation theory of type- A quivers, Yoneda evaluation, projective generators, Morita theory, mapping cones, and finite-space descent. The contribution here is the integrated boundary discipline:

- (i) high-dimensional source semantics are separated from their structured shadows by a purpose-faithful projection contract;
- (ii) finite observation is separated from finite proof by an explicit model of information, queries, encoding, and verification;
- (iii) persistence and finite-diagram mathematics are separated into two cores connected by typed interfaces;
- (iv) record globalization is separated from object globalization;
- (v) conclusion strength is target-relative and downward closed rather than silently scalarized;
- (vi) proof composition is non-compensating and failure-localizing;
- (vii) success theorems are paired with counterexamples that identify the load-bearing hypotheses.

This architecture is particularly useful when an argument combines numerical or persistence information with categorical reconstruction. The former is often robust but lossy; the latter is exact but requires actual maps, relations, and coherence. Treating them as one undifferentiated layer either discards useful robustness or introduces unsupported structure.

1.5. Scope and source convention. The default persistence scope is pointwise finite-dimensional constructible one-parameter persistence over a field. The default diagrammatic scope consists of finite small categories, finite-dimensional diagrams, and bounded complexes. Finite Alexandrov spaces and finite covers are used for strict descent; separately named model-certified derived descent may be admitted only through its own theorem.

No theorem below asserts a classification of wild multiparameter persistence, an arbitrary infinite-category result, unbounded-derived reconstruction, unrestricted stack or hypersheaf descent, or a named external open problem. The source corpus [16, 17, 18, 19, 20, 21] remains the canonical authority for theorem identity, migration, and complete audit records. The present paper is designed to be mathematically self-contained at the level of the stated kernel results, while Appendix A records the exact source map.

2. HIGH-DIMENSIONAL PROJECTION STRUCTURAL ANALYSIS

2.1. Purpose-faithful structural projection.

Definition 2.1 (AK structural projection datum). Let $\mathcal{X}$ be a declared source class and let $\mathcal{O}$ be a Core-readable target class. An *AK structural projection datum* for a source feature Θ_{src} and an observable feature Θ_{obs} consists of a route

$$\mathfrak{R}_{\text{AK}} : \mathcal{X} \rightsquigarrow \mathcal{O}, \quad X \longmapsto F_X,$$

together with the source and target semantics, variance, coefficient and parameter conventions, the direction of use, the exact source subclass, a theorem or controlled-defect statement relating $\Theta_{\text{src}}(X)$ and $\Theta_{\text{obs}}(F_X)$, and an explicit failure boundary. The arrow $\rightsquigarrow$ permits an objectwise or partially functorial construction. It is replaced by a functor arrow only when identities and composition are proved.

Definition 2.2 (Purpose-faithfulness). The datum in [Theorem 2.1](#) is *purpose-faithful* in the direction $\Theta_{\text{src}} \Rightarrow \Theta_{\text{obs}}$ if a theorem proves that implication on the declared source class. It is purpose-faithful in the reverse direction only if a separate theorem proves reflection. Purpose-faithfulness is therefore feature-relative and direction-relative; it is not categorical faithfulness, conservativity, full faithfulness, or equivalence unless those stronger properties are separately proved.

This formulation recovers the original meaning of “high-dimensional projection.” The source may have many degrees, parameters, local conditions, layers, or nonlocal compatibility constraints, while the shadow is chosen so that one declared structural feature becomes finitely certifiable. The theory does *not* currently assert a universal numerical dimension inequality such as $\dim F_X < \dim X$. The dimension reduction is structural and claim-relative, not an unproved topological-dimension theorem.

Proposition 2.3 (No theorem-bearing shadow without purpose-faithfulness). *A specified construction $X \rightsquigarrow F_X$ with no proved purpose-faithfulness statement yields only a candidate Core input. No source-domain implication may be inferred from a Core verdict on F_X .*

Proof. Without a preservation or reflection theorem, the source feature and the observed feature inhabit different predicates on different object classes. A verdict on the latter supplies no typed implication concerning the former. $\square$

2.2. Structured shadows and loss profiles. For a purpose-faithful projection, the first persistent shadow is

$$B_{n,W}(F_X) = W_W P_n(F_X),$$

and the repaired shadow is

$$\text{Eval}_{n,W,\tau}(F_X) = T_\tau^{\text{bar}}(B_{n,W}(F_X)).$$

These are not alternative names for X . They retain only the information proved to survive the realization, coefficient passage, parameter extraction, windowing, and thresholding steps.

Definition 2.4 (Projection loss profile). The *loss profile* of a structural projection records, for every downstream feature actually used, whether it is preserved, reflected, reconstructed only after an auxiliary theorem, controlled by a quantified defect, or forgotten. It also records the strongest permissible conclusion: predicate, invariant, object, morphism, exact, derived, or another explicitly named target strength.

Proposition 2.5 (Loss cannot be recovered by downstream syntax). *If two non-equivalent source objects have identical projected data at every field consumed by a route, then no downstream construction using only those fields can Return a source distinction unless a new rigidity theorem introduces additional information.*

Proof. The downstream route factors through the common projection. Therefore it assigns the same internal record to both source objects. A claimed exact source distinction would contradict this factorization unless an additional theorem restricts the source class or supplies new data. $\square$

2.3. Collapse as a typed structural analysis. The v21 theory does not reinstate a universal exact collapse functor. Instead, *Collapse* denotes a typed family of structural tests on the shadow. Depending on the declared route, these may include

- (i) vanishing of the repaired persistent shadow $\text{Eval}_{n,W,\tau}(F_X)$;
- (ii) finite detector certification of that vanishing or nonvanishing;
- (iii) a realized higher-Ext obstruction;
- (iv) kernel and cokernel profiles of an actual tower comparison morphism;
- (v) terminal-generic defect dimensions μ_{Collapse} and ν_{Collapse} ;
- (vi) long-transient and full-interior defect predicates;
- (vii) finite-state, Noetherian, or derived-limit reduction to an infinite target.

The source objects, stages, and conclusion strengths of these tests are different. The word *Collapse* does not identify them.

Definition 2.6 (Typed Collapse record). A *typed Collapse record* contains exactly the invoked stages among the list above, their actual source objects and morphisms, their comparison modes, thresholds, monitored degrees, finite-to-infinite mechanism when required, and their separate results. An absent actual comparison morphism makes a Type IV diagnostic undefined; it does not create a nonzero defect.

Proposition 2.7 (No automatic Collapse promotion). *For separately declared downstream objects A_X, B_X and an actual comparison morphism ϕ , none of the following implications is automatic:*

$$\begin{aligned} \text{Eval}_{n,W,\tau}(F_X) = 0 &\implies \text{Ext}^q(A_X, B_X) = 0, \\ (\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0) &\implies \ker \phi = \text{coker } \phi = 0, \\ \text{local Collapse zero} &\implies \text{global object zero}, \\ \text{Core Collapse zero} &\implies \Theta_{\text{src}}(X). \end{aligned}$$

Each implication requires its own realization, coverage, descent, interior-defect, rigidity, or Return theorem.

Proof. The antecedents and consequents are predicates on different source objects or at different information strengths. The counterexamples in Appendix B show that the missing transitions fail independently: scalar or barcode data need not determine morphisms, terminal zero need not remove interior defects, pairwise local data need not satisfy a cocycle, and a projected predicate need not reconstruct the source. $\square$

2.4. Projection–Return asymmetry.

Theorem 2.8 (Mathematical projection-to-Return composition). *Let X be a source object and suppose there is a theorem-bearing mathematical chain*

$$\begin{aligned} X &\xrightarrow{\text{proved typed Interface}} F_X = \mathfrak{R}_{\text{AK}}(X) \longrightarrow B_{n,W}(F_X) \\ &\longrightarrow \text{Eval}_{n,W,\tau}(F_X) \longrightarrow \mathbf{C}_{\text{fin}}(X) \longrightarrow \mathbf{C}_{\text{AK}}(X) \\ &\xrightarrow{\text{Return at strength } \sigma} E_{\sigma}(X). \end{aligned}$$

Assume that every mathematical arrow has a declared source and target, direction, hypotheses, and conclusion strength; every infinite quantifier is discharged by a named reduction; every invoked descent or promotion is justified by an effectivity or rigidity theorem; and the final arrow is a Return theorem at target-relative strength σ . Then the source-domain

conclusion $E_\sigma(X)$ follows. If any required mathematical Interface, certificate, reduction, descent, rigidity, or Return arrow is absent, the displayed mathematical composite is undefined at that target.

Proof. The positive assertion is the typed mathematical composition theorem proved in Theorem 10.11. The negative assertion follows because each required edge is a conjunct in the domain of that composition. An absent edge leaves the composite undefined rather than supplying a weaker implication automatically. $\square$

Proposition 2.9 (Source-controlled AK licensing boundary). *To register the conclusion of Theorem 2.8 as a theorem-bearing AK-HPDST v21 route, the canonical source statement additionally requires complete source binding, a benchmark-quarantine cut when applicable, a typed evidence graph, immutable theorem and artifact identities, and the Chapter 8 audit/manifest closure. Passing these predicates licenses the recorded AK use of the mathematical chain. It does not strengthen $E_\sigma(X)$, weaken its hypotheses, or create a missing mathematical arrow.*

Proof. Main Proposition 9.127 and Theorem 9.128 include a distinct audit/evidence-graph arrow between the finite or reduced certificate and the Core verdict. These clauses determine whether the v21 verifier may consume the printed mathematical chain under the claimed identities. The mathematical implication remains the one proved by the component theorems; identity and audit closure neither add a stronger target nor replace an absent component. $\square$

Remark 2.10 (The governing asymmetry). Projection reveals a structured shadow. Finite certificates may compress the proof of a shadow-level predicate. Diagrammatic reconstruction and descent recover only the structure licensed by their inputs. Return alone carries the certified statement home. AK does not identify the shadow with its source; it makes the theorem required for that identification explicit.

2.5. Mathematically distinguished results. The projection philosophy is supported by concrete theorems rather than by architecture alone. The following results are the mathematically most distinctive parts of the kernel, each with an exact scope and a visible non-claim boundary.

TABLE 1. Mathematical highlights and contribution classes.

Result	Contribution class	Exact mathematical content	Boundary
Sharp repaired Collapse	AK-local sharp theorem	Two-sided threshold-gap stability for hard deletion and exact zero separation on the τ -reduced image	No global non-expansiveness or exact collapse functor
Certificate complexity	AK-local model theorem	Finite Pareto frontier in $\mathbb{N}^4$; private-witness lower bounds; right-unbounded finite-detector no-go	Model-relative, not a lower bound for all proof systems
Rank reconstruction	Exact finite-model result	Finite Möbius inversion and coordinate-query minimum $N_m = (m + 1)(m + 2)/2$	The inversion is classical; the lower bound is for the declared coordinate model; morphisms require enriched data
Finite-state limit	Exact finite-state result	$\varprojlim V_n \cong \text{im } T^e$, $\varprojlim^1 V_n = 0$, and a finite apex test	Requires a typed finite-state presentation of the actual bonding maps
Diagrammatic globalization	Classical package with AK interfaces	Generator zero reflection, cone detection, Morita reconstruction, Alexandrov realization, and strict descent	Finite categories/posets, actual maps, cocycles, and effectivity
Causal blocker frontier	Appendix-local proof-DAG theorem	Required nodes of status reject or undefined have a unique earliest failure antichain	Certified mathematical obstruction is a separate output; the antichain does not prove the external target false

These results play different roles. The first two are repaired or model-relative AK-local contributions; the rank and finite-state results are exact theorems in explicit finite models; the globalization theorem is primarily a carefully typed package of classical categorical ingredients; and the causal-blocker theorem is a finite-poset statement about proof objects. This classification is part of the claim discipline: architectural integration is mathematically testable, but it is not presented as a new Morita, Yoneda, or sheaf-descent theory.

3. RELATION TO EXISTING LITERATURE AND MATHEMATICAL POSITIONING

3.1. Persistence modules, barcodes, and stability. Pointwise finite-dimensional one-parameter persistence modules decompose into interval modules under standard hypotheses, and the resulting barcode is unique up to permutation of the interval summands [6, 5, 12]. The interleaving and bottleneck viewpoints provide the basic stability language [10, 3]. The present paper does not replace those results. Its persistence contribution begins at a discontinuous operation that is not covered by naive functorial stability: hard deletion of all finite bars of length at most a fixed threshold. Because a small endpoint perturbation may move a bar across the threshold, hard deletion is neither globally non-expansive nor, in general, induced by a Serre localization. The relevant theorem is therefore a conditional transport result with a sharp two-sided exclusion band.

A second distinction concerns observations and certificates. A finite table of ranks or detector values is evidence only relative to a declared source class and a theorem proving soundness, completeness, or reconstruction. This is elementary in principle but becomes load-bearing once bounded windows, endpoint decorations, tail behavior, and adaptive queries are combined. The certificate model in [section 6](#) makes the information field, query model, encoding, and verifier explicit and keeps query count, bit length, verification work, and artifact size as separate cost coordinates.

3.2. Finite quivers, projective probes, and Morita reconstruction. Finite one-parameter persistence on an endpoint-complete grid is a representation of a linearly oriented type- A quiver. Interval representations, rank invariants, and Möbius inversion are standard manifestations of this finite-type representation theory [7, 1, 12]. The role of the persistence-to-diagram interface is not to reprove Gabriel’s classification in new notation, but to state exactly when a persistence object, rather than only its barcode, has been transported to an actual finite diagram with an action on morphisms.

For a finite category P , the category $\text{Fun}(P, \text{Vect}_k^{\text{fd}})$ is an abelian functor category. Linearized representables are projective by Yoneda, their finite direct sum is a projective generator, and classical Morita theory reconstructs the category from its endomorphism algebra [11, 1]. Passing to bounded derived categories uses standard homological algebra [8, 13]. What matters for the present architecture is the separation among three logically different statements: a generator may reflect zero, it may generate a perfect or bounded-derived subcategory under additional hypotheses, and its enriched Hom data may reconstruct objects and morphisms. None of these statements is interchangeable with a scalar rank test.

3.3. Finite spaces, sheaves, and descent. A finite poset determines an Alexandrov space, and sheaves on that space can be represented by diagrams with a variance convention fixed by the choice of upper or lower opens [2, 9, 14]. Sheaves satisfy effective descent for open covers, but a descent datum contains actual overlap isomorphisms and cocycle identities. Pairwise isomorphism classes or equal local invariants are not descent data. This distinction is the object-level counterpart of the finite-observation/finite-proof distinction in persistence.

The diagrammatic globalization theorem in [section 9](#) packages finite-category presentation, projective-generator probes, cone detection, Morita reconstruction, Alexandrov realization, and strict descent. Its scope is deliberately finite: finite categories or posets, finite-dimensional values, bounded complexes, and one explicitly selected descent model. General stacks, hypersheaves, microlocal categories, and unrestricted homotopy-coherent descent lie outside the stable theorem.

TABLE 2. Standing notation.

Symbol	Meaning
$\text{Pers}_{\mathbb{k}}^c$	pointwise finite-dimensional constructible one-parameter persistence modules over $\mathbb{k}$
T_τ^{bar}	hard deletion of finite bars of length at most τ ; infinite bars are retained
d_B	bottleneck distance with finite-to-diagonal cost $\ell(I)/2$ and the standard infinite-bar convention
Core_P	persistence-certificate core
Core_{DG}	finite-diagram and globalization core
$\mathfrak{R}_{\text{AK}}$	purpose-faithful AK structural projection from a source object to a Core-readable shadow
$\mu_{\text{Collapse}}, \nu_{\text{Collapse}}$	terminal-generic kernel and cokernel Collapse defect dimensions, when an actual comparison morphism is present
P2DG	typed persistence-to-diagram interface
$\text{Diag}_P(\mathbb{k})$	$\text{Fun}(P, \text{Vect}_{\mathbb{k}}^{\text{fd}})$ with actual natural transformations
G_P	finite sum of linearized covariant representables
Λ_P	$\text{End}(G_P)^{\text{op}}$
X_P	upper Alexandrov space associated with a finite poset P
Γ	propagated mathematical assumption context
S_T	downward-closed supported strength set for a fixed target T
Return_σ	target conclusion at declared strength σ

3.4. Position of the integrated framework. The paper’s integrated contribution is architectural but mathematically testable. Each passage

$$\begin{aligned} \text{source} &\longrightarrow \text{structured shadow} \longrightarrow \text{Collapse certificate} \\ &\longrightarrow \text{finite diagram} \longrightarrow \text{global object} \longrightarrow \text{typed Return}. \end{aligned}$$

has a separate source type, target type, theorem, strength ceiling, and counterexample boundary. The framework therefore differs both from a survey, which merely juxtaposes standard results, and from a universal metatheorem, which would obscure the hypotheses carried by each component. The synthesis theorems below are finite compositions of printed mathematical statements. Their novelty claim is limited to this typed, non-compensating composition and to the AK-local repaired and model-relative results identified in Appendix A.

4. STANDING NOTATION AND DEPENDENCY CONVENTION

Throughout, $\mathbb{k}$ is a field, $W \subseteq \mathbb{R}$ is a declared persistence window, $\tau > 0$ is a hard-deletion threshold, and P is a finite small category unless explicitly specialized to a finite poset. The principal notation is collected in table 2.

The four synthesis results have a strict dependency order. The repaired-certificate theorem supplies certified persistence information. The reconstruction theorem upgrades only those inputs satisfying the P2DG contract to actual finite diagrams. The globalization theorem consumes actual diagrams and descent data, not barcode classes. The Return theorem then composes the resulting target slice with source realization, any required

finite-to-infinite reduction, rigidity, and a final interpretation theorem. Thus

$$\text{Theorem A} \prec \text{Theorem B} \prec \text{Theorem C} \prec \text{Theorem D}$$

is a dependency statement, not a claim that every application invokes every theorem. A route may begin directly in Core_{DG} , may terminate at an internal predicate, or may omit descent when its target is already global.

5. TYPED ARCHITECTURE AND PROOF SEMANTICS

5.1. Persistence objects and finite diagrams. Fix a field $\mathbb{k}$. Let

$$\text{Pers}_{\mathbb{k}} := \text{Fun}((\mathbb{R}, \leq), \text{Vect}_{\mathbb{k}})$$

be the category of one-parameter persistence modules [5, 12]. We use the constructible pointwise finite-dimensional subcategory $\text{Pers}_{\mathbb{k}}^c$: on each bounded interval there is a finite critical set outside which all structure maps are isomorphisms, and critical values have no finite accumulation point.

For a finite small category P , write

$$\text{Diag}_P(\mathbb{k}) := \text{Fun}(P, \text{Vect}_{\mathbb{k}}^{\text{fd}}), \quad D_P^b(\mathbb{k}) := D^b(\text{Diag}_P(\mathbb{k})).$$

Morphisms in $\text{Diag}_P(\mathbb{k})$ are actual natural transformations. A graph with vector spaces and linear maps is not yet a P -diagram unless the identities, composition table, and relations of P are satisfied.

Definition 5.1 (Conclusion strengths). For a fixed target record, a route may support some of the following conclusion types:

predicate, numerical invariant, object class, object isomorphism,
morphism, exact structure, quasi-isomorphism, derived equivalence,
external Return.

The supported conclusions form a downward-closed subset of a declared target-relative preorder. No total order, lattice structure, or greatest element is assumed.

The definition prevents a common category error. Equality of scalar invariants may imply a predicate but not an object isomorphism. An object equivalence may fail to determine morphisms. A determinant or ideal may be an exact Return in its own target category while remaining weaker than module reconstruction.

5.2. The persistence-to-diagram interface.

Definition 5.2 (P2DG interface record). A persistence-to-diagram interface record is a tuple

$$\mathcal{I}_{P \rightarrow DG} = (\mathcal{C}_{\text{src}}, \mathcal{C}_{\text{tgt}}, F_{\text{obj}}, F_{\text{mor}}, \text{var}, \text{exact}, \text{reflect}, \text{info}, \text{loss}, \mathcal{A}),$$

where the entries specify the source and target categories, object and morphism action modes, variance, exactness profile, reflected predicates, retained information, discarded information, and immutable supporting artifacts.

The morphism field may be absent. In that case the interface is objectwise only and cannot supply a natural transformation, cone, or descent map. Similarly, an exact equivalence on an ambient finite-grid category does not make hard threshold deletion exact, and a reconstruction theorem on a linear quiver does not extend by relabelling to arbitrary finite categories.

5.3. Record globalization and object globalization. A certificate atlas may glue local records and establish a global record-level predicate. That is not the same as constructing a global mathematical object.

Definition 5.3 (Descent datum). Let X be a finite space with a finite open cover $\{U_i\}_{i \in I}$. A strict descent datum in a category of sheaves or bounded complexes consists of local objects E_i , actual isomorphisms

$$g_{ij} : E_j|_{U_i \cap U_j} \xrightarrow{\sim} E_i|_{U_i \cap U_j},$$

and the literal cocycle identities

$$g_{ij}g_{jk} = g_{ik} \quad \text{on } U_i \cap U_j \cap U_k.$$

A model-certified derived datum replaces literal equalities by specified coherent higher cells and includes a separately proved effectivity theorem.

Pairwise existence of isomorphisms is weaker than a descent datum. Even a complete list of pairwise maps does not imply the cocycle condition.

5.4. Proof DAGs and non-compensation.

Definition 5.4 (Typed proof DAG). A typed proof DAG is a directed acyclic graph whose vertices are theorem-bearing obligations, construction records, or target statements, and whose edges record source and target types, variance, assumptions, strength transport, and artifact identity. Countable dependence additionally requires a well-founded rank or a separately certified finite-to-infinite reduction.

Proposition 5.5 (Non-compensation). *Let a target route require vertices $v_1, \dots, v_r$. If one required vertex is missing, ill-typed, rejected, or obstructed, then passing values at the remaining vertices do not establish the target. In particular, metric reserve, redundant computations, successful compilation, replay success, or agreement among verifiers cannot replace an actual morphism, a descent datum, an effectivity theorem, a rigidity theorem, or a source interpretation.*

Proof. The target obligation is a typed conjunction. Each required vertex supplies a proposition in a specified hom-set, category, or target record. Evidence inhabiting a different field or codomain does not fill the missing slot. Without a conversion theorem there is no well-typed substitution, hence no derivation of the target. $\square$

5.5. Statuses and result sorts. The distinction between missing evidence and false evidence is mathematically relevant. We use four atomic statuses for theorem-facing obligations:

pass, reject, undefined, not_invoked.

Missing or ill-typed evidence gives `undefined`; a well-typed false obligation gives `reject`; and `not_invoked` requires a positive scope exclusion. At the final Bridge target, the result sort is

reject, undefined, obstructed, partial_return, regularized_return, return.

The identical words “reject” and “undefined” in different fields do not identify the sorts.

TABLE 3. Typed stages and forbidden promotions.

Stage	Admissible output	Not supplied automatically
Persistence evaluation	repaired profile, comparison bound, zero predicate	finite diagram, morphism, exactness
Finite certificate	sound and complete verdict in a declared model	representative, object descent, external theorem
P2DG	finite diagram object; actual morphisms only on a morphism route	arbitrary finite-category theorem, source reconstruction
Core-DG	typed generator, cone, Morita, or descent result	external Return, unselected-lane effectivity
Bridge	target-relative Return under propagated assumptions	stronger target, smaller assumption context

6. REPAIRED PERSISTENCE AND FINITE CERTIFICATES

6.1. Barcode conventions and hard threshold deletion. Every $M \in \text{Pers}_{\mathbb{k}}^c$ admits a locally finite interval decomposition on bounded ranges. We write $\mathcal{B}(M)$ for its barcode and use the standard bottleneck convention: finite bars may be matched to the diagonal at half their length, an infinite bar is never matched to the diagonal at finite cost, and two infinite bars $[a, \infty)$ and $[a', \infty)$ match at cost $|a - a'|$.

For a bar I define

$$\ell(I) = \begin{cases} b - a, & I \text{ has finite endpoints } a < b, \\ +\infty, & I \text{ is right-infinite.} \end{cases}$$

Finite-endpoint decoration does not affect $\ell(I)$.

Definition 6.1 (Hard threshold deletion). For $\tau > 0$ and a barcode B , define

$$T_{\tau}^{\text{bar}}(B) = \{I \in B : \ell(I) > \tau \text{ or } \ell(I) = +\infty\}.$$

Thus every finite bar of length at most τ is deleted and every infinite bar is retained.

The operation is idempotent and threshold-monotone:

$$T_{\tau}^{\text{bar}} T_{\tau}^{\text{bar}} B = T_{\tau}^{\text{bar}} B, \quad T_{\sigma}^{\text{bar}} T_{\tau}^{\text{bar}} B = T_{\max\{\sigma, \tau\}}^{\text{bar}} B.$$

It is an operation on barcode normal forms. On the full persistence category it is not asserted to be a strict exact endofunctor.

Proposition 6.2 (Positive-threshold short bars are not a Serre class). *For $\tau > 0$, the constructible persistence modules all of whose bars have finite length at most τ are not extension-closed in general.*

Proof. For $\tau = 1$ there is a short exact sequence

$$0 \longrightarrow \mathbb{k}_{[1,2)} \longrightarrow \mathbb{k}_{[0,2)} \longrightarrow \mathbb{k}_{[0,1)} \longrightarrow 0.$$

The subobject and quotient have length one, while the middle interval has length two. Rescaling gives the statement for arbitrary τ . $\square$

6.2. Failure of global non-expansiveness. Hard classification at the boundary $\ell = \tau$ is discontinuous unless both inputs avoid the boundary.

Example 6.3 (Global non-expansiveness fails). Fix $0 < \eta < \tau/4$ and set

$$B = \{[0, \tau + \eta)\}, \quad C = \{[0, \tau - \eta)\}.$$

Matching the two finite bars costs 2η , while deleting at threshold τ gives

$$T_\tau^{\text{bar}} B = B, \quad T_\tau^{\text{bar}} C = 0.$$

Hence

$$d_B(B, C) = 2\eta, \quad d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) = \frac{\tau + \eta}{2} > 2\eta.$$

Thus T_τ^{bar} is not globally non-expansive.

The example also explains why a one-sided reserve is insufficient. A small endpoint displacement may move a bar from the deleted side to the retained side. The correct theorem is two-sided.

6.3. Sharp threshold-gap stability. For a barcode B define its threshold clearance

$$\text{clear}_\tau(B) = \inf_{I \in B_{\text{fin}}} |\ell(I) - \tau|,$$

with value $+\infty$ when B has no finite bars. For a pair, set

$$\text{gap}_\tau(B, C) = \min\{\text{clear}_\tau(B), \text{clear}_\tau(C)\}.$$

Theorem 6.4 (Sharp two-sided threshold-gap stability). *Let B and C be finite barcode profiles, possibly with infinite bars. Let $\tau > 0$ and $\varepsilon \geq 0$. Assume*

$$d_B(B, C) \leq \varepsilon$$

and assume that no finite bar of either barcode has length in the closed band

$$[\tau - \varepsilon, \tau + \varepsilon].$$

Then

$$d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon.$$

The half-width ε is sharp for a two-sided closed exclusion band.

Proof. The case $\varepsilon = 0$ is immediate. Assume $\varepsilon > 0$ and choose a bottleneck matching of cost at most ε .

Let $I \in B$ be retained. If I is infinite, finite-cost matching forces it to match an infinite bar of C , which is retained. Suppose I is finite. The exclusion-band hypothesis and $\ell(I) > \tau$ imply

$$\ell(I) > \tau + \varepsilon.$$

If I is matched to the diagonal, retain that diagonal match. If it is matched to a finite bar $J \in C$, then the endpoint displacement bound gives

$$|\ell(I) - \ell(J)| \leq 2\varepsilon.$$

Were J deleted, then $\ell(J) \leq \tau$. Since C avoids the closed band, in fact $\ell(J) < \tau - \varepsilon$, so

$$\ell(I) - \ell(J) > 2\varepsilon,$$

a contradiction. Hence J is retained. The symmetric argument treats retained bars of C . Restricting the original matching, while retaining permitted diagonal matches, gives a matching of the thresholded barcodes of cost at most ε .

For sharpness, fix $0 < c < \varepsilon$ and choose $0 < \eta < \varepsilon - c$. Take centered bars of lengths

$$L_+ = \tau + c + \eta, \quad L_- = \tau - c - \eta.$$

Both avoid the smaller band $[\tau - c, \tau + c]$ and their bottleneck distance is $c + \eta < \varepsilon$, yet the first is retained and the second is deleted. For τ sufficiently large the distance from the retained bar to zero exceeds ε . Thus no smaller universal two-sided half-width suffices. $\square$

Corollary 6.5 (Clearance form). *If $d_B(B, C) \leq \varepsilon$ and*

$$\varepsilon < \text{gap}_\tau(B, C),$$

then $d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon$.

6.4. Zero separation on the reduced image.

Definition 6.6 (τ -reduced profile). A barcode profile P is τ -reduced if $T_\tau^{\text{bar}} P = P$. Equivalently, every finite bar has length strictly greater than τ .

For a finite barcode P , its distance from zero is

$$d_B(P, 0) = \begin{cases} 0, & P = 0, \\ \frac{1}{2} \max_{I \in P} \ell(I), & P \neq 0 \text{ and all bars are finite,} \\ +\infty, & P \text{ contains an infinite bar.} \end{cases}$$

Theorem 6.7 (Sharp zero separation). *Let P, Q be finite τ -reduced barcode profiles, possibly with infinite bars. If*

$$d_B(P, Q) \leq \varepsilon, \quad 2\varepsilon \leq \tau,$$

then

$$P = 0 \iff Q = 0.$$

The constant is sharp.

Proof. Assume $P = 0$. If Q contains an infinite bar, then $d_B(P, Q) = +\infty$. If Q is nonzero and finite, every bar has length $> \tau \geq 2\varepsilon$, so

$$d_B(0, Q) = \frac{1}{2} \max_{I \in Q} \ell(I) > \varepsilon,$$

again a contradiction. Hence $Q = 0$. The converse is symmetric.

If $2\varepsilon > \tau$, choose L with $\tau < L \leq 2\varepsilon$ and let Q consist of one finite bar of length L . Then Q is τ -reduced and $d_B(0, Q) = L/2 \leq \varepsilon$, while $Q \neq 0$. $\square$

The mathematically sharp boundary is closed. In numerical or scalarized implementations one may impose the stricter reserve $2\varepsilon < \tau$ to avoid equality-sensitive rounding.

Corollary 6.8 (Metric-safe repaired-zero transport). *Under the hypotheses of [Theorem 6.4](#), if additionally $2\varepsilon \leq \tau$, then*

$$T_\tau^{\text{bar}} B = 0 \iff T_\tau^{\text{bar}} C = 0.$$

Proof. The thresholded profiles are τ -reduced. Apply [Theorems 6.4](#) and [6.7](#). $\square$

The conclusion concerns the exact zero predicate. It does not assert stability of arbitrary pointwise ranks or of a raw detector vector. A detector verdict may be transported only after its own soundness and completeness theorem identifies it with the certified zero predicate.

6.5. Window-first evaluation. Let F be a finite-type filtered complex and $P_i(F)$ its i th persistent homology. For a right-open window $W = [u, u')$ define the pre-threshold profile

$$B_{i,W}(F) = W_W P_i(F)$$

and the repaired evaluation

$$\text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(B_{i,W}(F)). \quad (2)$$

The order in (2) is part of the definition. Cropping and hard deletion do not commute automatically, especially when a globally infinite bar becomes finite after clipping. Any alternate order requires an exact commutation theorem for the declared window and endpoint policy.

6.6. Certificate models.

Definition 6.9 (Certificate model). A certificate model is a quadruple

$$\mathfrak{M} = (I, Q, E, V),$$

where I specifies information and advice available before queries, Q specifies the admissible access pattern, E specifies finite encoding and cost accounting, and V specifies the verifier and acceptance semantics.

For a certificate C_X of an input X , define the cost vector

$$\text{cost}_{\mathfrak{M}}(C_X) = (q, b, t, a) \in \mathbb{N}^4,$$

where q is the charged query count, b the encoded bit length, t the verifier cost in the declared model, and a the additional artifact footprint. For a family-relative scheme Π , one fixed verifier rule must be sound and complete for every input in the family; the scheme is not designed after unrestricted inspection of the unknown object.

Definition 6.10 (Model-relative minimal certificates). For a family $\mathcal{F}$ and predicate Θ , let

$$\mathcal{C}_{\mathcal{F}}^{\mathfrak{M},\text{fin}}(\Theta) \subseteq \mathbb{N}^4$$

be the achievable finite family-cost vectors of sound-and-complete schemes. Define

$$\text{MinCert}_{\mathcal{F}}^{\mathfrak{M}}(\Theta) = \text{ParetoMin}(\mathcal{C}_{\mathcal{F}}^{\mathfrak{M},\text{fin}}(\Theta)).$$

If no finite-cost scheme exists, the set is empty. Coordinate lower envelopes are defined by infima, with value $+\infty$ when the coordinate is never finite.

Proposition 6.11 (Finite Pareto frontier). *If $\mathcal{C}_{\mathcal{F}}^{\mathfrak{M},\text{fin}}(\Theta)$ is nonempty, then its Pareto frontier is a nonempty finite set, and every achievable vector dominates a frontier element.*

Proof. The coordinatewise order on $\mathbb{N}^4$ is a well-quasi-order by Dickson's lemma [15]. Hence every nonempty subset has finitely many minimal elements. Descending within the finite coordinate box below a fixed achievable vector reaches such a minimal element. $\square$

The set-valued frontier is essential. A scheme may trade fewer queries for longer answers, more verifier work, or larger proof artifacts. Without a declared utility function there is no canonical scalar cost.

6.7. A common-model exact benchmark. Fix rational $\tau > 0$, a bounded rational domain $W_\tau = [a, c)$, and pairwise disjoint nonempty rational half-open cells

$$U_j = [p_j, p_j + e_j) \subseteq W_\tau, \quad j = 1, \dots, m.$$

Let

$$B_0 = 0, \quad B_j = \{[p_j, p_j + \tau + e_j)\}, \quad \mathcal{F}_{\text{priv}} = \{B_0, B_1, \dots, B_m\}.$$

In the deterministic exact point-stencil model, a query at $t \in W_\tau$ returns

$$\dim_{\mathbb{K}} \text{im}(B(t) \longrightarrow B(t + \tau)).$$

The information field contains the family specification and the promise of family membership but not the identity of the unknown barcode.

Theorem 6.12 (Exact private-cell query complexity). *For the repaired-zero predicate*

$$\Theta(B) = 0 \iff T_\tau^{\text{bar}} B = 0,$$

the exact deterministic worst-case query complexity on $\mathcal{F}_{\text{priv}}$ is m , even for adaptive algorithms.

Proof. Choose one rational point $s_j \in U_j$ and query all m cells. The zero barcode returns zero at every point; B_j returns a nonzero image exactly on its active-start cell U_j . This gives a sound-and-complete nonadaptive scheme with $q = m$.

For the lower bound, use B_0 as a baseline and B_j as a private witness. Outside U_j , every answer for B_j agrees with the answer for B_0 . If an algorithm omits U_j along the zero transcript, it receives the same transcript on B_j and therefore produces the same verdict. Since the U_j are pairwise disjoint, every sound-and-complete deterministic algorithm must query each cell. Thus $q \geq m$. $\square$

Under finite encodings of the rational cells and finite or finitely coded coefficient data, the exhibited scheme has a finite full cost vector. The theorem does not apply to a model with hidden full-object access or uncharged object-dependent advice.

6.8. Right-unbounded no-go theorem. Boundedness or a separately certified tail theorem is not cosmetic. Without it, no uniformly complete finite stencil need exist.

Theorem 6.13 (Right-unbounded finite-detector no-go). *Let $W = [a, \infty)$, $\tau > 0$, $e > 0$, and*

$$\mathcal{U}_{a,\tau,e} = \{B_p = \{[p, p + \tau + e)\} : p \geq a\}.$$

For every finite stencil $T \subset W$ there is $B_p \in \mathcal{U}_{a,\tau,e}$ such that all queried structure-map images vanish while $T_\tau^{\text{bar}} B_p \neq 0$. Moreover, no deterministic oracle-closed scheme that halts after finitely many queries on every input is sound and complete for the repaired-zero predicate on $\{0\} \cup \mathcal{U}_{a,\tau,e}$.

Proof. A finite stencil is bounded above. Choose $p > \max T$. The active-start interval $[p, p + e)$ is disjoint from T , so all stencil values coincide with those of the zero barcode, while the bar length $\tau + e$ exceeds the threshold.

For an adaptive scheme, run the algorithm on zero. Its finite transcript queries a bounded set. Choose p strictly to the right of every queried point. Every answer on B_p equals the zero answer along that transcript; oracle-closedness forces the same subsequent queries and the same verdict. The algorithm cannot be correct on both inputs. $\square$

The no-go theorem is escaped only by additional proved structure, such as a known finite birth set, a global birth bound, eventual zero, eventual stabilization, a finite-to-infinite reduction, or a separately certified cofinal detector. A long observed plateau proves none of these conditions.

6.9. Composition and evidence reuse. Two finite certificate schemes may be composed when they use compatible families and models, their predicates and source stages are typed, the verifier contains a rule for the composite predicate, and their dependencies form an acyclic subgraph. If $c_1, c_2 \in \mathbb{N}^4$ are family costs and h_{12} is the composition overhead, then

$$\text{cost}(\Pi_1 \otimes \Pi_2) \preceq c_1 + c_2 + h_{12}.$$

At an object X , content-addressed query reuse refines the query coordinate to

$$|Q_1(X) \cup Q_2(X) \cup H_{12}(X)|.$$

Exact savings in bit length, verifier time, or artifact footprint require declared reuse rules. This local calculus is not a theorem that global certificate complexity equals the sum of local minima; overlap coherence, object descent, shared encodings, and global artifacts remain independent costs.

Theorem 6.14 (Repaired finite certificate synthesis). *Within a declared constructible one-parameter family, fixed window/endpoint convention, threshold τ , and certificate model $\mathfrak{M}$, the following hold.*

- (a) *Hard deletion admits metric transport under the sharp two-sided threshold-gap condition of Theorem 6.4.*
- (b) *On the τ -reduced image, $2\varepsilon \leq \tau$ gives exact zero transport.*
- (c) *On bounded promised families, sound-and-complete finite certificates may have exact model-relative upper and lower bounds, as in Theorem 6.12.*
- (d) *On the right-unbounded translation family of Theorem 6.13, no finite deterministic exact certificate scheme exists.*
- (e) *Composable certificates admit typed evidence reuse, but no local cost identity is promoted to a global descent or Return theorem.*

Proof. Items (a) and (b) are Theorems 6.4, 6.7 and 6.8. Item (c) is Theorem 6.12; item (d) is Theorem 6.13. Item (e) follows from content-addressed union of the queried evidence and the non-compensation principle of Theorem 5.5. $\square$

Remark 6.15 (Source). The canonical repaired-persistence and certificate statements are Main Theorems 2.46, 2.53, 2.56, 3.86, 3.93, and 3.97, with expanded proofs in Foundation Appendices A–B [16, 17].

7. FINITE-TO-INFINITE REDUCTION AND RECORD-LEVEL GLOBALIZATION

The word “finite” in a certificate theorem concerns the evidence available to a verifier. It does not imply that the mathematical target is finite. When the target contains an infinite tail, a limit, or an independently declared apex, a separate reduction theorem must identify the infinite object from finite data. This section records the two reductions used later: theorem-specific tail certification and record-level atlas globalization. Neither is object descent.

7.1. Tower variance and the finite-prefix boundary. A tower is not determined by dimensions, ranks, barcode classes, or a finite prefix. Its variance and actual transition maps are part of the input. We distinguish direct systems

$$V_0 \longrightarrow V_1 \longrightarrow V_2 \longrightarrow \cdots$$

from inverse systems

$$V_0 \longleftarrow V_1 \longleftarrow V_2 \longleftarrow \cdots$$

Reversing the displayed arrows changes the universal construction and is not a notational operation.

Proposition 7.1 (Finite-prefix non-determination). *For every $N \geq 0$ there are two inverse systems of finite-dimensional vector spaces whose objects and transition maps agree through level N but whose inverse limits are non-isomorphic. Consequently, agreement on a finite prefix does not prove eventual constancy, a limit formula, or apex agreement.*

Proof. Take the common prefix $V_0 = \cdots = V_N = \mathbb{k}$ with identity transitions. Extend it first by the constant identity tower. Extend it second by setting $V_{N+1} = 0$ and all later terms equal to zero. The first inverse limit is $\mathbb{k}$, whereas the second is zero. The two systems have identical finite data through level N . $\square$

The proposition is the tail analogue of the right-unbounded detector no-go theorem. In both cases an adversary places the distinguishing information outside the inspected region. A positive theorem must therefore provide a tail mechanism: eventual isomorphisms, finite length, a Noetherian image argument in the correct variance, a Mittag–Leffler condition, a finite-state recurrence, or another named reduction.

7.2. Finite-state inverse limits. We state a representative exact lane. A *typed finite-state inverse tower* is an inverse system of finite-dimensional $\mathbb{k}$ -vector spaces together with a finite state presentation that reproduces every object and transition map. The first repeated based state determines an entry index N and a period $r > 0$. Writing the transitions as $d_n : V_{n+1} \rightarrow V_n$, define the periodic monodromy

$$T = d_N d_{N+1} \cdots d_{N+r-1} : V_N \longrightarrow V_N.$$

Let $e = \dim_{\mathbb{k}} V_N$ and set

$$W_{\text{st}} := \text{im } T^e = \bigcap_{q \geq 0} \text{im } T^q.$$

Finite-dimensional image stabilization makes $T|_{W_{\text{st}}}$ an automorphism.

Theorem 7.2 (Exact finite-state inverse-limit certificate). *For a typed finite-state inverse tower as above, evaluation at the periodic entry induces an isomorphism*

$$\text{ev}_N : \varprojlim_n V_n \xrightarrow{\sim} W_{\text{st}}, \quad \varprojlim_n^1 V_n = 0.$$

The first isomorphism is canonical relative to the chosen entry-period record and the based state identifications. It is natural only for morphisms of finite-state towers commuting with those identifications and all transition matrices. Moreover,

$$\varprojlim_n V_n = 0 \iff T^e = 0 \iff T \text{ is nilpotent.}$$

Proof. For a compatible sequence $x = (x_n)$, periodic compatibility implies $x_N = T^q x_{N+qr}$ for every $q \geq 0$. Thus x_N lies in every image of T^q , hence in W_{st} .

Conversely, let $w \in W_{\text{st}}$. Since T restricts to an automorphism of W_{st} , define the period-boundary values by

$$x_{N+qr} = T^{-q} w.$$

The actual transition maps determine every intermediate coordinate inside each period, and composition determines the finite prefix. This gives a compatible sequence whose N -th coordinate is w . The two constructions are inverse.

The images of the transition maps stabilize periodically on the recurrent sector, so the inverse system is Mittag–Leffler. For a countable inverse system of vector spaces, the

standard Roos description then gives $\varprojlim^1 = 0$ [13, Sec. 3.5]. Finally, $W_{\text{st}} = 0$ exactly when $\text{im } T^e = 0$, which in finite dimension is equivalent to nilpotence of T . $\square$

Corollary 7.3 (Finite apex-agreement test). *Let A be a finite-dimensional declared apex with compatible maps $\alpha_n : A \rightarrow V_n$. Under Theorem 7.2, the induced map*

$$\alpha : A \longrightarrow \varprojlim_n V_n$$

is an isomorphism if and only if the entry component

$$\alpha_N : A \longrightarrow W_{\text{st}}$$

is an isomorphism.

Proof. Under the evaluation isomorphism of Theorem 7.2, the composite $A \rightarrow \varprojlim V_n \rightarrow W_{\text{st}}$ is precisely α_N . $\square$

Example 7.4 (A nontrivial stable-image limit). Let every term be $V_n = \mathbb{k}^3$ and every transition be

$$T = \text{diag}(1, 1, 0).$$

Then $e = 3$ and

$$W_{\text{st}} = \text{im } T^3 = \text{span}\{e_1, e_2\}.$$

Hence $\varprojlim V_n \cong \mathbb{k}^2$ and $\varprojlim^1 V_n = 0$, although the transition map is not an isomorphism on the whole three-dimensional space. Taking $A = \mathbb{k}^2$ with the standard inclusion into the first two coordinates verifies apex agreement by one finite matrix check.

The example is stronger than eventual constancy of the whole tower: the inverse limit retains the recurrent stable-image sector and discards the nilpotent sector. The strength is still model-relative. A finite automaton for dimensions or isomorphism classes is insufficient unless it determines the typed bonding data used to form T .

7.3. Finite-to-infinite soundness and independent defects.

Definition 7.5 (Finite-to-infinite reduction record). A finite-to-infinite reduction record specifies the tower variance and actual maps; a finite witness; the named reduction theorem and all hypotheses; the exceptional finite range and complete tail coverage; every ordinary-limit, derived-limit, stable-image, or recurrence clause used; the declared apex and actual comparison map; a downward-closed set of supported internal conclusion strengths; and explicit failure routes.

The internal strengths include, but are not limited to,

tail predicate, inverse-limit object, derived-limit clearance,
internal isomorphism, apex agreement.

Derived-limit clearance and apex agreement are independent: $\varprojlim^1 = 0$ does not identify an independently declared apex, while apex agreement does not erase a derived obstruction in a different exact sequence.

Theorem 7.6 (Finite-to-infinite soundness). *If a reduction record is complete and consistent, its finite witness passes, every hypothesis and tail obligation of the selected theorem passes, and every required limit, derived-limit, and apex-comparison clause passes, then every strength in the registered supported set holds. No strength outside that set follows. Missing required evidence makes the requested conclusion undefined; an evidenced false hypothesis rejects the selected lane.*

Proof. Apply the named mathematical reduction theorem to the verified tower and tail data. The record contributes typing and dependency control but no additional mathematical implication. Therefore its conclusion is exactly the supported set of the selected route. The status statements follow from the distinction between missing and false obligations. $\square$

7.4. Certificate atlases and global record zero. Finite evidence is also assembled across windows. Let a global persistence profile $B_{i,X}$ be related to local profiles B_i^α on windows W_α by exact, threshold-gap-safe, or theorem-controlled zero-reflecting interfaces. A certified local detector atlas records a complete detector on every local profile. The additional global obligation is retained-bar coverage: every globally retained interval must contain a certified local witness span of length greater than τ in at least one window.

Theorem 7.7 (Detector-atlas local-to-global zero). *Assume a declared global profile interface, complete local detector certificates, safe local zero transport, and certified global retained-bar coverage. Then*

$$T_\tau^{\text{bar}}(B_{i,X}) = 0 \iff D_\alpha(B_i^\alpha) = 0 \text{ for every } \alpha.$$

Proof. If the global repaired profile is zero, every exact restriction is repaired-zero; the local interface and detector soundness give local zero. Conversely, suppose all local detectors vanish and a globally retained interval I exists. Coverage supplies a window and a witness t with $t, t + \tau \in I \cap W_\alpha$. The exact restricted global profile is then nonzero after thresholding. Zero reflection across the local interface and detector completeness force a nonzero local detector value, a contradiction. $\square$

Coverage is load-bearing. If $X = [0, 2\tau)$ carries the bar $[0, 2\tau)$ and is covered by windows whose clipped bars all have length at most τ , every local repaired profile is zero although the global repaired profile is nonzero. Exact overlap coherence among the zero local records does not repair the missing global span.

Proposition 7.8 (Record globalization is not object descent). *A global record-zero verdict does not imply the existence or vanishing of a global object in a target category. Such a conclusion additionally requires actual local objects, transition equivalences, cocycle or higher coherence data, and an effective descent theorem that constructs the global object.*

Proof. A record atlas may contain only predicates, numerical bounds, or comparison certificates. These fields do not type an object of the target category, an overlap morphism, or a cocycle. Without those data the purported global object is not defined, so object-level vanishing is not a well-typed consequence. $\square$

The section supplies the finite evidence needed for infinite or global quantifiers without collapsing distinct targets. Tail reduction produces an internal limit or apex statement. Atlas coverage produces a global predicate about a declared profile. Object construction remains the responsibility of the diagrammatic descent theory in [section 9](#).

8. FINITE DIAGRAM RECONSTRUCTION

8.1. Endpoint-complete atom grids. A finite sampling grid is reconstructive only when it records the endpoint geometry that distinguishes constructible intervals. Let $W = [u, u')$ be a bounded right-open window and let

$$\Gamma = \{\gamma_0 < \gamma_1 < \cdots < \gamma_m\}$$

be a finite ordered probe set associated with an atom decomposition

$$\mathcal{A} = (A_0 < A_1 < \cdots < A_m).$$

The atoms include singleton critical values when endpoint decoration is to be recovered and the adjacent open atoms on which all structure maps are isomorphisms. A separate terminal tag records whether a window-cofinal summand is interpreted only inside W , as the clip of a bounded global interval, or as the clip of a right-infinite interval. That tag is input metadata; it is not inferred from the bounded restriction.

Definition 8.1 (Γ -constructible module). A persistence module P on W is Γ -constructible if it is constant up to canonical isomorphism on every atom A_i , every structure map inside an atom is an isomorphism, and all changes occur across the ordered atom boundaries encoded by Γ .

Let $\text{Rep}_{\mathbb{k}}(A_{[m]})$ denote representations of the linearly oriented quiver

$$0 \longrightarrow 1 \longrightarrow \cdots \longrightarrow m.$$

Restriction at the chosen probes defines

$$\text{Res}_{\Gamma} : \text{Pers}_{\mathbb{k}}^{\Gamma}(W) \longrightarrow \text{Rep}_{\mathbb{k}}(A_{[m]}).$$

Conversely, a quiver representation extends to a step persistence module by making it constant on each atom and using the quiver arrows across adjacent atoms. Denote this functor by Step_{Γ} .

Theorem 8.2 (Finite-grid encoding equivalence). *The functors*

$$\text{Res}_{\Gamma} : \text{Pers}_{\mathbb{k}}^{\Gamma}(W) \rightleftarrows \text{Rep}_{\mathbb{k}}(A_{[m]}) : \text{Step}_{\Gamma}$$

are quasi-inverse exact equivalences. In particular, restriction is full, faithful, conservative, and zero-reflecting on the declared Γ -constructible category.

Proof. By construction, $\text{Res}_{\Gamma} \text{Step}_{\Gamma}$ is the identity. For a Γ -constructible module P , within-atom transport gives a natural isomorphism

$$\text{Step}_{\Gamma} \text{Res}_{\Gamma}(P) \xrightarrow{\sim} P.$$

A natural transformation of step modules is determined by its probe components, and any natural transformation of quiver representations extends atomwise, proving fullness and faithfulness. Pointwise kernels and cokernels of maps between Γ -constructible modules remain Γ -constructible; restriction and step extension preserve them. Thus the equivalence is exact. Conservativity and zero reflection follow. $\square$

Remark 8.3 (Ambient equivalence versus repaired image). The theorem concerns the ambient Γ -constructible category. If the input is a repaired profile $P = T_{\tau}^{\text{bar}} B$, then every finite geometric interval has length $> \tau$. Its quiver image therefore lies in a generally proper full essential image. The ambient exact equivalence is not an exactness theorem for hard deletion.

If only a barcode isomorphism class is available, one may form the direct sum of the corresponding interval representations. That construction is lossless on objects when the atom dictionary and terminal policy are complete, but it is objectwise only until a morphism action is supplied.

8.2. Endpoint-complete reconstruction.

Theorem 8.4 (Endpoint-complete reconstruction). *Let P be a finite Γ -constructible persistence module. Assume that the atom record includes all finite geometric endpoints, singleton endpoint atoms, adjacent open atoms, endpoint decorations, and the terminal policy. Then P is reconstructed up to persistence-module isomorphism from $\text{Res}_{\Gamma} P$. Each*

barcode interval corresponds bijectively, with multiplicity, to an interval representation supported on a consecutive node set of $A_{[m]}$.

Proof. The module reconstruction is [Theorem 8.2](#). An interval summand is constant on each atom and its support is a consecutive sequence of atoms, hence restriction gives an indecomposable interval representation. Conversely, the union of a consecutive sequence of atoms is a decorated geometric interval. Singleton critical atoms determine inclusion or exclusion of finite endpoints. The quiver reconstructs the decorated interval inside W ; any distinction between bounded continuation and an infinite tail is supplied only by the declared terminal tag. Krull–Schmidt uniqueness preserves multiplicities. $\square$

An incomplete grid may compute all displayed matrices exactly and still fail to reconstruct. Omission of a critical value, merger of a singleton endpoint atom with an adjacent open atom, or omission of the terminal convention can identify nonisomorphic persistence modules.

8.3. Rank tables and finite Möbius inversion. For $V \in \text{Rep}_{\mathbb{k}}(A_{[m]})$ and $0 \leq i \leq j \leq m$, define

$$r_V(i, j) = \text{rank}(V_i \longrightarrow V_j),$$

where the map is the unique composite. Adopt boundary conventions

$$r_V(-1, j) = 0, \quad r_V(i, m+1) = 0, \quad r_V(-1, m+1) = 0.$$

Theorem 8.5 (Finite Möbius multiplicity formula). *Let $\mu_V(i, j)$ be the multiplicity of the interval representation supported on $[i, j]$. Then*

$$\mu_V(i, j) = r_V(i, j) - r_V(i-1, j) - r_V(i, j+1) + r_V(i-1, j+1). \quad (3)$$

The values $\mu_V(i, j)$ are nonnegative and determine V up to isomorphism.

Proof. Write

$$V \cong \bigoplus_{a \leq b} \mathbb{k}_{[a,b]}^{\oplus \mu_V(a,b)}.$$

An interval $[a, b]$ contributes one to $r_V(i, j)$ exactly when $a \leq i \leq j \leq b$. Hence

$$r_V(i, j) = \sum_{a \leq i, b \geq j} \mu_V(a, b).$$

This is the zeta transform on the product of the two finite chains. Applying the product Möbius function gives the second difference (3). Nonnegativity follows because the output is an interval multiplicity, and type- A Krull–Schmidt uniqueness gives reconstruction. $\square$

Corollary 8.6 (Decorated window-barcode Return). *Under the hypotheses of [Theorem 8.4](#), the complete rank table and atom dictionary reconstruct the complete decorated barcode of P on W . A declared terminal tag may append a certified global-tail interpretation, but the bounded rank table alone does not determine that interpretation.*

This is an object-level Return. It does not reconstruct a filtered chain complex, a chain map, a chosen derived realization, or an external-domain object.

8.4. Enriched probes and morphism reconstruction. Scalar rank data are sufficient for object classification in the finite type- A family, but they do not carry arbitrary morphisms. To recover the diagram functorially, one retains the projective probe system.

For $0 \leq i \leq m$, let P_i be the indecomposable projective supported on $[i, m]$, and let S_i be the simple at i . For $i < m$ there is a canonical projective resolution

$$0 \longrightarrow P_{i+1} \longrightarrow P_i \longrightarrow S_i \longrightarrow 0.$$

Proposition 8.7 (Local projective–Ext sequence). *For $V \in \text{Rep}_{\mathbb{k}}(A_{[m]})$ and $i < m$, there is a natural exact sequence*

$$0 \longrightarrow \text{Hom}(S_i, V) \longrightarrow V_i \longrightarrow V_{i+1} \longrightarrow \text{Ext}^1(S_i, V) \longrightarrow 0.$$

Moreover,

$$\text{Hom}(P_i, V) \cong V_i, \quad \text{Ext}^p(-, -) = 0 \quad (p \geq 2)$$

in this hereditary quiver category.

Proof. Apply $\text{Hom}(-, V)$ to the displayed projective resolution. Yoneda evaluation gives $\text{Hom}(P_i, V) \cong V_i$, and the induced map is the quiver arrow $V_i \rightarrow V_{i+1}$. The linearly oriented type- A path algebra is hereditary. $\square$

Theorem 8.8 (Enriched reconstruction). *The collection of spaces $\text{Hom}(P_i, V)$ together with the precomposition maps induced by $P_{i+1} \rightarrow P_i$ reconstructs V functorially. The simple Hom and Ext^1 terms are natural kernel/cokernel consistency data. Their dimensions alone are not a complete morphism signature.*

Proof. Yoneda identifies $\text{Hom}(P_i, V)$ with V_i . Under this identification, precomposition by $P_{i+1} \rightarrow P_i$ is the arrow $V_i \rightarrow V_{i+1}$. Hence every object and arrow is recovered. Forgetting the actual spaces and maps can identify distinct diagrams with the same scalar dimensions. $\square$

The distinction between the object branch and the morphism branch is a structural firewall. A quiver-morphism `Return` is admissible only when an actual persistence morphism is restricted to a common endpoint-complete grid or when a separate theorem constructs the required natural transformation.

8.5. P2DG handoff.

Theorem 8.9 (P2DG handoff). *On the ambient endpoint-complete Γ -constructible category, the P2DG interface*

$$P \longmapsto \text{Res}_{\Gamma} P$$

may be consumed at exact-equivalence strength on objects and actual morphisms. On the repaired-profile essential image it remains full, faithful, conservative, zero-reflecting, and lossless at persistence-module strength, with exactness restricted to ambient exact sequences whose relevant terms, kernels, and cokernels stay in the declared subcategory. A barcode-only branch returns only an objectwise decorated-barcode-equivalent finite object and supplies no cone or descent-map data.

Proof. The actual-module branch is [Theorems 8.2](#) and [8.4](#). The repaired image is a full essential image determined by the support-length predicate. The barcode-only construction has no action on morphisms; therefore any operation requiring an actual map is type-correct only on the actual-module branch or after an additional morphism theorem. $\square$

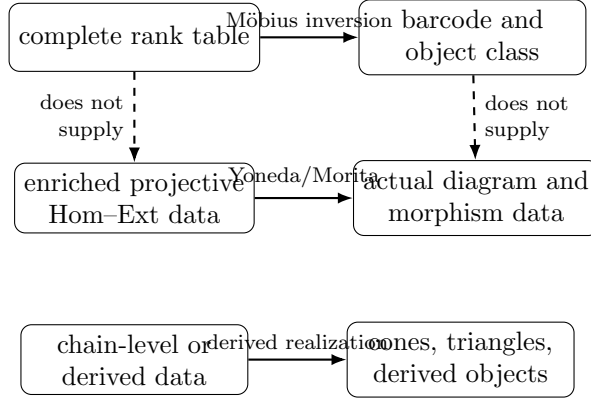


FIGURE 2. Information-strength ladder for the P2DG interface. Scalar rank data reconstruct the object class in the finite type- A regime; actual morphisms and derived structures require enriched or chain-level inputs.

8.6. Exact coordinate-query complexity. Let

$$N_m = \#\{(i, j) : 0 \leq i \leq j \leq m\} = \frac{(m+1)(m+2)}{2}.$$

Let $\mathcal{F}_m$ be the family of all finite type- $A_{[m]}$ representations with arbitrary nonnegative interval multiplicities. In the deterministic coordinate rank-query model, a query selects one pair (i, j) and returns $r_V(i, j)$. Query choice may depend on earlier answers; object-dependent advice, arbitrary linear combinations, and encoded scalars are excluded.

Theorem 8.10 (Exact rank-query complexity). *Exact reconstruction of every object in $\mathcal{F}_m$ requires and is achieved by exactly N_m coordinate rank queries.*

Proof. The complete rank table has N_m entries and reconstructs all multiplicities by Theorem 8.5, proving the upper bound.

For the lower bound, write the rank vector as

$$r = A\mu,$$

where $\mu \in \mathbb{Z}_{\geq 0}^{N_m}$ is the interval-multiplicity vector and

$$A_{(i,j),(a,b)} = 1 \iff a \leq i \leq j \leq b.$$

Consider a deterministic algorithm using fewer than N_m coordinate queries. For every proper coordinate subset Q , the restricted matrix A_Q has a nonzero rational kernel; after clearing denominators choose $0 \neq h_Q \in \ker(A_Q) \cap \mathbb{Z}^{N_m}$. There are finitely many such Q , so

$$H = \max_Q \|h_Q\|_\infty < \infty.$$

Choose $M > H$ and run the algorithm on the constant multiplicity vector $M\mathbf{1}$. Let Q be the coordinates queried along that transcript. The distinct vectors

$$M\mathbf{1}, \quad M\mathbf{1} + h_Q$$

are both nonnegative and give identical answers on every coordinate in Q . Inductively the adaptive algorithm follows the same query path and returns the same output on both inputs, so it cannot be exact. Hence at least N_m queries are necessary. $\square$

If every interval multiplicity is at most B , then every rank is at most $N_m B$. A complete certificate therefore has

$$q = N_m, \quad b = O(N_m \log(N_m B + 1)), \quad t_{\text{arith}} = O(N_m),$$

for direct second-difference inversion, apart from source-matrix checks and artifact costs. The query lower bound is model-relative: an oracle returning the full barcode in one answer is a different information model.

Theorem 8.11 (Finite diagram reconstruction synthesis). *For endpoint-complete constructible one-parameter persistence on a bounded window:*

- (a) *restriction to the atom grid is an exact equivalence with a finite linear-quiver category;*
- (b) *the complete rank table reconstructs the decorated barcode and object isomorphism class by finite Möbius inversion;*
- (c) *the enriched projective probe reconstructs the finite diagram functorially;*
- (d) *actual morphisms, cones, exact structures, and derived objects require the corresponding enriched or chain-level data;*
- (e) *in the coordinate rank-query model, N_m is the exact query complexity for the unrestricted interval-multiplicity family.*

Proof. Combine [Theorems 8.2, 8.4, 8.5 and 8.8 to 8.10](#). □

Remark 8.12 (Source). The finite-grid, Möbius, and P2DG statements are canonically owned by Main Chapter 4 and expanded in Foundation Appendix C [\[16, 17\]](#).

9. FINITE DIAGRAMMATIC GLOBALIZATION

9.1. Finite category presentations. A finite category is not determined by its underlying directed graph. The input consumed by the diagrammatic core contains the finite object set, every finite hom-set, identity morphisms, the complete composition table, all relations among generators, and the chosen variance. If the opposite category is used, it is declared explicitly.

Example 9.1 (Relations are reconstruction data). Consider vertices $0, 1, 2$ and arrows

$$a : 0 \rightarrow 1, \quad b : 1 \rightarrow 2, \quad c : 0 \rightarrow 2.$$

Let P_1 impose the relation $c = ba$, and let P_2 retain c as an independent arrow. A graph representation with matrices (A, B, C) is a P_1 -diagram exactly when $C = BA$, whereas every triple is a P_2 -diagram. Taking $A = B = 0$ and $C = \text{id}_{\mathbb{k}}$ gives a representation of P_2 but not of P_1 . Thus an artifact storing only the graph and arrow matrices is incomplete unless it also stores and verifies the relation table.

For a finite poset $(P, \leq)$ we use the associated category with a unique arrow $p \rightarrow q$ when $p \leq q$. Passing to P^{op} reverses every nonidentity arrow and is not a notation-only change.

9.2. The abelian finite-diagram category.

Proposition 9.2 (Objectwise abelian structure). *For every finite small category P , the category $\text{Diag}_P(\mathbb{k})$ is a $\mathbb{k}$ -linear abelian category. Kernels, cokernels, images, coimages, finite limits, and finite colimits are computed objectwise. A sequence is exact exactly when it is exact at every object of P .*

Proof. Functor categories into an abelian category are abelian, with limits and colimits computed pointwise. In the present finite-dimensional setting every pointwise kernel and cokernel remains finite-dimensional. Naturality of the induced structure maps follows from the universal properties of kernels and cokernels. □

The statement “finite” does not imply semisimple. The finite-dimensional category algebra may have infinite global dimension; this distinction will matter when zero detection is compared with thick generation.

9.3. Representable projectives and the finite generator. For $p \in P$, define the linearized covariant representable

$$G_p(q) = \mathbb{k}[P(p, q)].$$

Postcomposition gives the diagram structure.

Proposition 9.3 (Finite Yoneda evaluation). *For every $E \in \text{Diag}_P(\mathbb{k})$, evaluation at id_p gives a natural isomorphism*

$$\text{Hom}_{\text{Diag}_P(\mathbb{k})}(G_p, E) \xrightarrow{\sim} E(p).$$

Consequently G_p is projective.

Proof. Given $x \in E(p)$, define a natural transformation $\eta^x : G_p \rightarrow E$ by

$$\eta_q^x([u : p \rightarrow q]) = E(u)x.$$

This is inverse to evaluation at $[\text{id}_p]$. Since evaluation at p is exact by [Theorem 9.2](#), $\text{Hom}(G_p, -)$ is exact. $\square$

Define the finite projective generator

$$G_P = \bigoplus_{p \in P} G_p.$$

Proposition 9.4 (Generator property). *The object G_P is a projective generator. In particular,*

$$\text{Hom}(G_P, E) = 0 \iff E = 0.$$

Every finite-dimensional diagram is a quotient of a finite direct sum of representables.

Proof. By [Theorem 9.3](#),

$$\text{Hom}(G_P, E) \cong \bigoplus_{p \in P} E(p),$$

so the Hom space vanishes exactly when every component vanishes. Choosing bases of the spaces $E(p)$ and the associated Yoneda maps gives an objectwise surjection from a finite sum of representables. $\square$

9.4. Derived zero reflection and cone detection.

Theorem 9.5 (Generator zero reflection). *For every $E \in D_P^b(\mathbb{k})$,*

$$E \simeq 0 \iff \text{RHom}(G_P, E) \simeq 0.$$

Because G_P is projective, the derived probe is represented, for every bounded representative, by the ordinary complex

$$\text{Hom}(G_P, E^\bullet).$$

Proof. Regard G_P as a complex concentrated in degree zero. It is K -projective. Exactness of $\text{Hom}(G_P, -)$ yields

$$H^r \text{RHom}(G_P, E) \cong \text{Hom}(G_P, H^r(E)).$$

If the probe is acyclic, [Theorem 9.4](#) gives $H^r(E) = 0$ for every r , so $E \simeq 0$. The converse is immediate. $\square$

Zero reflection must not be confused with thick generation of all bounded derived objects. If the algebra associated with P has infinite global dimension, G_P still detects zero in D^b , while $\text{thick}(G_P)$ may be only the perfect subcategory.

Theorem 9.6 (Cone-isomorphism criterion). *For a morphism $f : E \rightarrow E'$ in $D_P^b(\mathbb{k})$,
 f is an isomorphism $\iff \text{Cone}(f) \simeq 0$.*

Consequently,

$$f \text{ is an isomorphism} \iff \text{RHom}(G_P, \text{Cone}(f)) \simeq 0.$$

Proof. The first equivalence is the standard cone criterion in a triangulated category. The second follows from [Theorem 9.5](#). $\square$

A concrete cone requires an actual chain map, including source and target identities, degree and sign conventions, and naturality checks. A bottleneck matching, rank equality, or objectwise correspondence does not define f .

Example 9.7 (Objectwise isomorphism is not naturality). For the arrow category $0 \rightarrow 1$, consider

$$E = (\mathbb{k} \xrightarrow{\text{id}} \mathbb{k}), \quad F = (\mathbb{k} \xrightarrow{0} \mathbb{k}).$$

The values at both vertices are isomorphic. A natural isomorphism would require invertible scalars s_0, s_1 satisfying

$$s_1 \text{id} = 0 s_0,$$

which is impossible. Objectwise isomorphism classes do not reconstruct a diagram isomorphism.

9.5. Finite Morita reconstruction. Set

$$\Lambda_P = \text{End}_{\text{Diag}_P(\mathbb{k})}(G_P)^{\text{op}}.$$

The opposite convention makes $\text{Hom}(G_P, E)$ a left Λ_P -module by precomposition. It also makes G_P a right Λ_P -object, so $G_P \otimes_{\Lambda_P} M$ is defined for a left module M .

For each p , let $e_p \in \Lambda_P$ be the primitive idempotent corresponding to id_p . For an arrow $\alpha : p \rightarrow q$, let λ_α be the corresponding category-algebra element. If M is a left Λ_P -module, define

$$E_M(p) = e_p M, \quad E_M(\alpha)(m) = \lambda_\alpha m.$$

The algebra relations guarantee functoriality.

Theorem 9.8 (Finite Morita reconstruction). *The functors*

$$\text{Hom}(G_P, -) : \text{Diag}_P(\mathbb{k}) \longrightarrow \Lambda_P\text{-mod}_{\text{fd}}$$

and

$$G_P \otimes_{\Lambda_P} - : \Lambda_P\text{-mod}_{\text{fd}} \longrightarrow \text{Diag}_P(\mathbb{k})$$

are mutually quasi-inverse exact equivalences. Their termwise extensions induce an exact triangulated equivalence

$$D_P^b(\mathbb{k}) \xrightarrow{\text{RHom}(G_P, -)} D^b(\Lambda_P\text{-mod}_{\text{fd}}).$$

Proof. For $E \in \text{Diag}_P(\mathbb{k})$,

$$e_p \text{Hom}(G_P, E) \cong \text{Hom}(G_p, E) \cong E(p).$$

Under this identification, the action of λ_α is precisely the structure map $E(\alpha)$. Thus the explicit construction $M \mapsto E_M$ is a quasi-inverse on objects and morphisms. Exactness follows from the projectivity of G_P and objectwise exactness. Applying the exact equivalences degree-wise gives the bounded-derived equivalence. $\square$

$$\begin{array}{ccc}
& \text{Hom}(G_P, -) & \\
\text{Diag}_P(\mathbb{k}) & \xrightleftharpoons{\quad} & \Lambda_P\text{-mod}_{\text{fd}} \\
& G_P \otimes_{\Lambda_P} - & \\
\uparrow p \mapsto \mathcal{E}(U_p) & \text{Alexandrov} & \\
\text{Sh}_{\text{fd}}(X_P; \mathbb{k}) & \xrightarrow[\text{effective finite descent}]{} & \text{global sheaf or bounded complex}
\end{array}$$

FIGURE 3. Finite diagrammatic globalization. Morita and Alexandrov realizations are exact equivalences at their declared levels; the bottom arrow additionally requires actual overlap maps, cocycles, and an effectivity theorem.

The enriched module action is essential. A list of dimensions or arrow ranks is a complete reconstruction invariant only on a separately proved restricted family. For a category with independently labelled arrows, two nonconjugate action tuples may have identical individual ranks.

9.6. Finite Alexandrov realization. Let P be a finite poset. Its upper Alexandrov topology consists of upper sets. The minimal open neighbourhood of p is

$$U_p = \{q \in P : p \leq q\}.$$

If $p \leq q$, then $U_q \subseteq U_p$, and sheaf restriction gives a map

$$\mathcal{E}(U_p) \longrightarrow \mathcal{E}(U_q),$$

so the upper-sheaf model is covariant in P .

Theorem 9.9 (Finite Alexandrov realization). *For a finite poset P , there is an exact equivalence*

$$\text{Sh}_{\text{fd}}(X_P; \mathbb{k}) \xrightarrow{\sim} \text{Diag}_P(\mathbb{k}), \quad \mathcal{E} \longmapsto (p \mapsto \mathcal{E}(U_p)).$$

A quasi-inverse sends a diagram E to the sheaf

$$\tilde{E}(U) = \left\{ (s_p)_{p \in U} \in \prod_{p \in U} E(p) : E(p \leq q)s_p = s_q \text{ for all } p \leq q \text{ in } U \right\}.$$

The equivalence extends to bounded derived categories.

Proof. Restriction along $U_q \subseteq U_p$ gives the diagram associated with a sheaf. For a diagram, the compatible-family formula defines a sheaf because compatible local families glue pointwise and uniquely. Evaluation at p identifies $\tilde{E}(U_p)$ with $E(p)$. Conversely, a sheaf on a finite Alexandrov space is determined by its values on minimal opens, since every open is their union. Exactness is stalkwise and corresponds to objectwise exactness. The derived statement follows by applying the exact equivalence degreewise. $\square$

Lower opens, cosheaves, contravariant diagrams, and P^{op} are separate models. Variance must be part of the interface record.

9.7. Effective strict descent.

Theorem 9.10 (Effective finite Alexandrov descent). *Let $X_P = \bigcup_{i=1}^N U_i$ be a finite cover by upper Alexandrov opens. Every strict descent datum*

$$(\mathcal{E}_i, g_{ij})$$

with actual local sheaves, actual overlap isomorphisms, and literal cocycle identities is effective. There exists a sheaf $\mathcal{E}$ on X_P , unique up to unique isomorphism compatible with the datum, and $\mathcal{E}|_{U_i} \cong \mathcal{E}_i$. Compatible local morphisms descend uniquely.

Proof. For an open $U \subseteq X_P$, define

$$\mathcal{E}(U) = \left\{ (s_i)_i \in \prod_i \mathcal{E}_i(U \cap U_i) : s_i|_{U \cap U_i \cap U_j} = g_{ij}(s_j|_{U \cap U_i \cap U_j}) \right\}.$$

Restriction is componentwise. To verify the sheaf axiom, glue each local component using the sheaf axiom for $\mathcal{E}_i$; the pairwise identities hold locally on the cover and hence globally by uniqueness. Projection to the i th component gives $\mathcal{E}|_{U_i} \rightarrow \mathcal{E}_i$, whose inverse is constructed using the transition maps. The cocycle ensures compatibility. Uniqueness and descent of morphisms follow by the same local gluing argument. $\square$

Example 9.11 (Pairwise isomorphism without a cocycle). Suppose three members of a cover have a common nonempty triple intersection and all local objects are the one-dimensional constant sheaf. Choose

$$g_{12} = \text{id}, \quad g_{23} = \text{id}, \quad g_{13} = -\text{id}$$

over a field of characteristic different from two. Every pair is isomorphic, but

$$g_{12}g_{23} \neq g_{13}.$$

The datum is not a strict descent datum and [Theorem 9.10](#) does not apply.

9.8. Bounded-complex descent.

Theorem 9.12 (Strict bounded-complex descent). *Let $K_i^\bullet$ be bounded complexes of finite-dimensional sheaves on the members of a finite Alexandrov cover. Assume a common finite amplitude, actual chain isomorphisms on overlaps, and literal cocycle identities. Then the datum descends degreewise to a bounded global complex $K^\bullet$, unique up to unique chain isomorphism compatible with the local data. Compatible chain maps and chain homotopies descend.*

Proof. Apply [Theorem 9.10](#) in each degree. Compatibility of the local differentials with the transition maps makes them descend to global differentials. The equation $d^{n+1}d^n = 0$ holds locally and hence globally. The common amplitude preserves boundedness. The same local-to-global argument descends chain maps, homotopies, and their defining equations. $\square$

A complete higher-coherence record in another enhancement is not automatically effective. Such a route is admitted only when it selects one model, supplies all required higher cells, and names a theorem proving effectivity in that model.

9.9. Global zero and isomorphism certificates.

Proposition 9.13 (Joint conservativity). *For a finite open cover $X_P = \bigcup_i U_i$, the exact restriction functors*

$$D^b(\text{Sh}_{\text{fd}}(X_P; \mathbb{k})) \longrightarrow D^b(\text{Sh}_{\text{fd}}(U_i; \mathbb{k}))$$

are jointly conservative.

Proof. If every restriction of E is zero, then every cohomology sheaf restricts to zero. Each point lies in some U_i , so every stalk of every cohomology sheaf vanishes. Hence $E \simeq 0$. $\square$

Theorem 9.14 (Global zero certificate). *Let E be a global bounded derived sheaf, given directly or constructed by an admitted descent lane. Let G_i be the finite projective generator of the local diagram category on U_i . Then*

$$E \simeq 0 \iff \mathrm{RHom}(G_i, E|_{U_i}) \simeq 0 \text{ for every } i.$$

Proof. The forward implication is functorial. Conversely, local generator zero reflection gives $E|_{U_i} \simeq 0$ for every i , and [Theorem 9.13](#) gives $E \simeq 0$. $\square$

Theorem 9.15 (Global isomorphism certificate). *Let $f : E \rightarrow E'$ be an actual global morphism, or the global morphism obtained by effective descent of compatible local morphisms. Then*

$$f \text{ is an isomorphism} \iff \mathrm{RHom}(G_i, \mathrm{Cone}(f|_{U_i})) \simeq 0 \text{ for every } i.$$

Proof. Restriction is exact and commutes with mapping cones. Local probe vanishing implies that every restricted cone is zero. Joint conservativity makes the global cone zero, and [Theorem 9.6](#) gives the conclusion. $\square$

Effectivity alone is not enough for global zero detection. A global construction must be paired with exact restriction and joint conservativity. For example, the projection

$$\mathrm{Vect}_{\mathbb{k}}^{\mathrm{fd}} \times \mathrm{Vect}_{\mathbb{k}}^{\mathrm{fd}} \longrightarrow \mathrm{Vect}_{\mathbb{k}}^{\mathrm{fd}}$$

admits the chosen lift $V \mapsto (V, 0)$ but is not conservative.

Theorem 9.16 (Finite diagrammatic globalization synthesis). *Within the finite diagrammatic scope, the following chain is valid:*

$$\begin{aligned} \text{actual finite diagram} &\longrightarrow \text{enriched generator module} \longrightarrow \text{cone and Morita control} \\ &\longrightarrow \text{Alexandrov realization} \longrightarrow \text{effective finite globalization.} \end{aligned}$$

More precisely:

- (a) $\mathrm{Diag}_P(\mathbb{k})$ is abelian with objectwise exactness;
- (b) G_P is a finite projective zero-reflecting generator;
- (c) actual morphisms are isomorphisms exactly when their cones are zero;
- (d) the enriched generator probe gives exact abelian and bounded-derived Morita reconstruction;
- (e) finite-poset diagrams are exactly finite-dimensional sheaves on the upper Alexandrov space;
- (f) strict sheaf and bounded-complex descent are effective;
- (g) local generator and cone certificates yield global zero and isomorphism only after global existence and joint conservativity.

No external-domain theorem follows without later Bridge and Return nodes.

Remark 9.17 (Source). The canonical finite-diagram and globalization statements are Main Theorems 7.14, 7.16, 7.21, 7.27, 7.31, 7.34, 7.42, 7.43, and 7.54, with expanded proofs in Appendix DG [16, 18].

Proof. Combine [Theorems 9.2, 9.4 to 9.6, 9.8 to 9.10, 9.12, 9.14 and 9.15](#). $\square$

10. TYPED BRIDGE COMPOSITION AND RETURN

10.1. Universal Bridge data. An internal certificate has no external semantics until a source realization identifies the external datum with a typed internal object or predicate. Conversely, an external theorem whose hypotheses have not been connected to the internal

output supplies no Return. We formalize the complete route rather than a slogan about correspondence.

Definition 10.1 (Universal Bridge datum). A Bridge datum is

$$\mathfrak{B} = (\mathcal{S}, \mathcal{R}, \mathcal{C}, \mathcal{T}, \Sigma, \Gamma, \mathfrak{D}, \mathcal{A}),$$

where:

- (i) $\mathcal{S}$ is the exact source semantics;
- (ii) $\mathcal{R}$ is a source realization, including variance and coefficient conventions;
- (iii) $\mathcal{C}$ is the selected Core/transfer/globalization route;
- (iv) $\mathcal{T}$ is the exact target record;
- (v) Σ is the supported target-relative strength set;
- (vi) Γ is the propagated assumption context;
- (vii) $\mathfrak{D}$ is the typed defect and failure record;
- (viii) $\mathcal{A}$ is the immutable theorem, source, construction, and replay bundle.

Two Bridge data are composable only when the displayed codomain and domain, variance, coefficients, target identity, and assumptions agree or are connected by a proved conversion theorem.

Definition 10.2 (Typed component). A component is a theorem-backed transformation

$$C_i : (\mathcal{X}_i, \Gamma_i, S_i) \longrightarrow (\mathcal{Y}_i, \Gamma'_i, S'_i),$$

with an exact source type, target type, assumptions, supported lower set, failure leaves, and artifacts. A component may be exact, metric-gap-certified, theorem-controlled nonmetric, or not compared; no mode implies another.

Typical components include source realization, coefficient or base change, homological compression, persistence reconstruction, finite-diagram reconstruction, spectral-sequence transfer, finite-state or Noetherian reduction, derived-limit control, descent, duality, coherence, rigidity, and final interpretation. Co-location in one document does not create an interface between components.

10.2. Assumption and strength propagation. Let $D = (V, E)$ be a finite typed route. For a target vertex t , write $D_{\leq t}$ for the induced sub-DAG on t and all its ancestors.

Definition 10.3 (Route context). The propagated assumption context of a complete route is

$$\Gamma(D_{\leq t}) = \bigcup_{v \in V(D_{\leq t})} \Gamma_v \cup \bigcup_{e \in E(D_{\leq t})} \Gamma_e,$$

modulo only those assumption reductions established by explicit theorems.

Definition 10.4 (Route-supported strength). If S_v and S_e are the downward-closed supported strength sets of the necessary vertices and edges, define

$$S(D_{\leq t}) = \bigcap_{v \in V_{\text{req}}} S_v \cap \bigcap_{e \in E_{\text{req}}} S_e.$$

The intersection is taken only after all sets have been interpreted in the same target-relative preorder.

A route can therefore become weaker without becoming invalid. For example, exact object reconstruction may be unavailable while a vanishing predicate or determinant remains valid. Such a result is not a failed exact theorem rewritten rhetorically; it is a separate target-relative Return.

10.3. Partial and regularized Returns.

Definition 10.5 (Partial Return). A partial Return names an exact quotient of information, such as a predicate, invariant, truncation, graded object, semisimplification, stable class, determinant, ideal, or pseudo-isomorphism class. It records the target functor Q , target equivalence relation, residual defect, and theorem preventing promotion to the full object.

Theorem 10.6 (Partial Return principle). *Suppose every necessary component passes for a target functor*

$$Q : \mathcal{C}_{\text{target}} \longrightarrow \mathcal{Q},$$

but the route lacks rigidity for objects of $\mathcal{C}_{\text{target}}$. Then the route may Return $Q(Y)$ at the certified invariant strength and may not Return Y .

Proof. The route proves the statement in $\mathcal{Q}$. Recovering Y requires a reverse implication or conservativity theorem for Q on the declared source class. In its absence, distinct target objects may have identical images, so exact object Return would be unsound. $\square$

Definition 10.7 (Regularized target). A regularized target record contains a new target identifier T_{reg} , a theorem from the audited internal conclusion to T_{reg} , its relation to the original target, every singular or correction datum, and an explicit statement that the original exact target has not been returned.

A valid regularized theorem produces `regularized_return` for the replacement target. It does not convert a rejected or undefined original target into a pass.

10.4. Rigidity and promotion.

Definition 10.8 (Rigidity package). A rigidity package names the observation or reconstruction functor, its essential source class, the external equivalence relation, a finite presentation or generator theorem, normalization data, and a theorem promoting observed equality or internal equivalence to the requested external strength.

Proposition 10.9 (No promotion without rigidity). *If an observation functor identifies two non-equivalent source objects and no rigidity theorem excludes that pair on the declared class, then no downstream route consuming only that observation can Return exact source objects or morphisms.*

Proof. Every downstream construction factors through the same observation. The two source objects remain indistinguishable at every later node, contradicting exact reconstruction if such a Return were asserted. $\square$

This proposition is the general form of the objectwise-versus-natural and predicate-versus-source boundaries already encountered.

10.5. Infinite quantifiers and tail selection. An infinite target requires a theorem that reduces its quantifier to finite data. A finite prefix, apparent plateau, or state summary is not itself such a theorem. A valid route selects one mechanism, for example:

- eventual constancy with an explicit stabilization witness;
- finite-state periodicity or finite monodromy;
- Noetherian stabilization;
- a Mittag-Leffler or derived-limit theorem;
- a source-specific recurrence or compactness theorem.

The selected mechanism must return the exact infinite object or predicate consumed downstream. Vanishing of a derived-limit defect, for example, does not by itself identify an external apex.

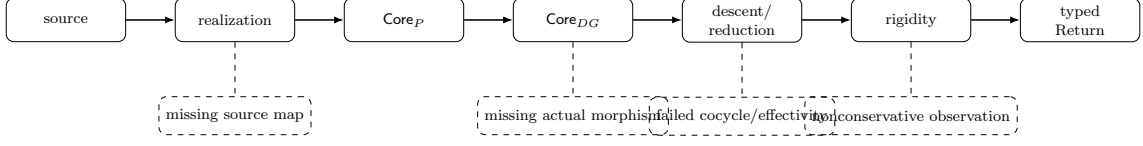


FIGURE 4. A target-sliced Bridge DAG. Dashed leaves are causal blockers; success elsewhere does not compensate for a missing required edge.

10.6. The unified defect stack. For a complete route, write

$$\mathfrak{D}_{\text{Bridge}} = (D_{\text{src}}, D_{\text{real}}, D_{\text{coeff}}, D_P, D_{DG}, D_{HT}, D_{\text{tail}}, D_{\text{desc}}, D_{\text{coh}}, D_{\text{rig}}, D_{\text{repl}}, D_{\text{return}}, D_{\text{cert}}).$$

Each coordinate is a typed family of failure leaves. A missing source realization is not a nonzero mathematical obstruction; it is an undefined edge. A proved nonzero terminal defect is an obstruction only after the corresponding comparison object and actual morphism exist. Causal ordering prevents the same failure from being counted simultaneously as two primary defects.

Definition 10.10 (Primary Bridge result). For a fixed target record, the primary result is exactly one of

reject, undefined, obstructed, partial_return, regularized_return, return.

10.7. Unified closure.

Theorem 10.11 (Typed end-to-end Return). *Let D be a finite composable mathematical Bridge DAG with source $\mathcal{S}$, target $\mathcal{T}$, requested assumption context Γ , and requested strength σ . Assume:*

- (1) *a proved typed source realization is defined, including its purpose-faithfulness direction, variance, coefficient conventions, and loss profile;*
- (2) *every invoked persistence-core and diagrammatic-core node is proved at the consumed type;*
- (3) *every transfer, coefficient, base-change, or reusable component is proved at the consumed type;*
- (4) *every infinite quantifier is discharged by a theorem-certified finite-to-infinite, recurrence, compactness, or derived-limit reduction;*
- (5) *every overlap, higher-cell, and parallel-route coherence obligation required by the selected route is satisfied;*
- (6) *every invoked descent or global construction is effective in one declared lane;*
- (7) *every promotion beyond observation strength is justified by a rigidity or conservativity theorem on the declared source class;*
- (8) *a final interpretation theorem maps the proved internal conclusion to the exact target record;*
- (9) *every propagated mathematical assumption is contained in Γ ;*
- (10) *σ lies in the supported lower set of every necessary vertex and edge after all strengths are interpreted in the same target-relative preorder.*

Then

$$\Gamma \vdash \text{Return}_{\sigma}(\mathcal{S}, \mathcal{T}).$$

Every registered weakening of σ is also available. This route licenses neither a stronger conclusion nor a strictly smaller assumption context.

Proof. Evaluate the target slice in a topological order. At each edge, type-correct composition transports the preceding conclusion only to the printed codomain of the next theorem. The reduction, coherence, descent, and rigidity hypotheses discharge the corresponding nonlocal obligations. The final interpretation theorem then produces the exact target statement. Assumptions propagate by union except where an explicit theorem removes them, and the conclusion remains inside the common supported lower set. Hence the composite derivation is valid at (Γ, σ) . $\square$

Proposition 10.12 (Audit data are non-promotional but licensing-relevant). *Version identifiers, exact source bindings, quarantine records, hashes, compilation, artifact closure, and semantic replay may certify which theorem-bearing route and artifact have been used, but they do not strengthen the mathematical conclusion of Theorem 10.11. Conversely, failure of a required identity or audit check prevents the source-controlled v21 verifier from licensing that recorded route, without by itself proving the underlying printed mathematical statement false.*

Proof. The mathematical conclusion depends on the theorem hypotheses and their derivation. Identity, quarantine, and replay predicates determine admissible source-controlled consumption of that derivation. The two roles are distinct: no workflow predicate creates a stronger mathematical target, while an unverified identity cannot support the claimed source-bound invocation. $\square$

Corollary 10.13 (Canonical v21 licensing form). *If, in addition to the mathematical hypotheses of Theorem 10.11, source binding and benchmark quarantine pass and the artifact, semantic-replay, and immutable-manifest clauses of Main Theorem 9.47 pass, then the source-controlled primary result for the exact target is return. These added clauses change the admissibility of the recorded route, not its target-relative mathematical strength.*

Remark 10.14 (Source). The canonical unified closure theorem is Main Theorem 9.47, with research-spine and component expansions in Appendices HT and UB [16, 19, 20].

Corollary 10.15 (Non-composition). *The existence of an internal certificate and an external theorem does not by itself imply an external Return. In general, source realization, internal transfer, finite-to-infinite or descent reduction, rigidity when needed, and source-domain interpretation are independent necessary arrows.*

Proof. The arrows have different source and target types. Omitting any arrow leaves no composable path from the source record to the requested target record. $\square$

10.8. Alternative routes and failure isolation. Alternative complete routes to the same target contribute the downward closure of the union of their supported strength sets. An incomplete route contributes no target strength, and fragments of two incomplete routes cannot be spliced unless a new theorem-backed route explicitly composes them. For independent targets, the combined result lies in a product category and retains coordinatewise strengths; it is not an untyped intersection.

Theorem 10.16 (Unique causal blocker antichain). *Let G be a finite acyclic proof DAG and let F be the set of required nodes whose atomic status is reject or undefined. Define*

$$B(G, F) = \{v \in F : \text{no } u \in F \text{ satisfies } u <_G v\},$$

where $u <_G v$ means that a nontrivial directed path runs from u to v . Then $B(G, F)$ is the unique inclusion-minimal antichain contained in F such that every node of F lies on or downstream from an element of $B(G, F)$. Changing only strict descendants of unchanged blocker nodes cannot repair the route.

TABLE 4. Typical Bridge outcomes.

Result	Mathematical meaning	Typical cause
undefined	requested proposition is not evaluable	missing theorem, map, source binding, or ill-typed datum
reject	a well-typed obligation is false	failed equation, false hypothesis, verifier counterexample
obstructed	a certified nonzero obstruction blocks the route	nonzero defect, nontrivial cone, nonzero descent or coherence obstruction class
partial_return	a weaker exact quotient is returned	observation complete but rigidity absent
regularized_return	a replacement target is returned	singular or exceptional original target
return	exact target returned at declared strength	every necessary node and edge passes

Proof. Every $v \in F$ has a $<_G$ -minimal failed or undefined ancestor because F is finite. These minimal elements are pairwise incomparable and every failed or undefined node lies downstream from one of them. Any subset of F with the same ancestry-covering property must contain each minimal element, proving uniqueness and inclusion-minimality. Altering strict descendants leaves the atomic status of each unchanged blocker untouched. $\square$

A certified nonzero obstruction is not an element of F merely because it prevents an exact Return: it is a theorem-bearing output with primary result *obstructed*. The blocker antichain instead localizes missing or false obligations, such as a missing source realization, absent actual morphism, rejected tail theorem, failed cocycle equation, missing effectivity, or absent final Return theorem.

11. KERNEL DEMONSTRATIONS AND EXACT COMPLEXITY

This section gives finite examples that exercise the three nouns in the title: certificates, diagrammatic globalization, and typed Returns. The examples are deliberately small enough for direct replay, but each is paired with a boundary statement that blocks an invalid promotion.

11.1. A complete rank certificate on $A_{[3]}$. Consider the representation

$$V_0 = \mathbb{k}^2, \quad V_1 = \mathbb{k}^3, \quad V_2 = \mathbb{k}^3, \quad V_3 = \mathbb{k},$$

with structure matrices

$$f_{01} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f_{12} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad f_{23} = (1 \ 0 \ 0).$$

The complete rank table is

$$\begin{array}{c|cccc}
 r(i, j) & j = 0 & j = 1 & j = 2 & j = 3 \\
 \hline
 i = 0 & 2 & 2 & 1 & 1 \\
 i = 1 & & 3 & 2 & 1 \\
 i = 2 & & & 3 & 1 \\
 i = 3 & & & & 1
 \end{array} \quad (4)$$

Applying (3) gives

$$\mu(0, 1) = \mu(0, 3) = \mu(1, 2) = \mu(2, 2) = 1,$$

with all other multiplicities zero.

Proposition 11.1 (Exact reconstruction of the example). *The representation is reconstructed up to isomorphism as*

$$V \cong \mathbb{k}_{[0,3]} \oplus \mathbb{k}_{[0,1]} \oplus \mathbb{k}_{[1,2]} \oplus \mathbb{k}_{[2,2]}.$$

If V is the atom restriction of an endpoint-complete persistence module, the corresponding decorated barcode is reconstructed exactly.

Proof. The Möbius multiplicities computed from (4) produce the displayed direct sum. Its vertex dimensions are $(2, 3, 3, 1)$, and its composite ranks agree with the table. Uniqueness of the interval decomposition gives the object isomorphism. Endpoint-complete step reconstruction then returns the persistence module and decorated intervals. $\square$

The certificate uses all ten coordinates

$$N_3 = \frac{(3+1)(3+2)}{2} = 10.$$

By Theorem 8.10, no deterministic adaptive algorithm using fewer coordinate rank queries reconstructs the entire unrestricted multiplicity family. The example is therefore not only sufficient but query-minimal in the declared model.

11.2. Why scalar probes are not enriched reconstruction. Let P have two objects p, q and two parallel arrows $a, b : p \rightrightarrows q$. Over a field of characteristic different from two, define diagrams E and E' by

$$\begin{aligned}
 E(p) = E(q) = E'(p) = E'(q) &= \mathbb{k}^2, \\
 E(a) = E(b) &= I_2, \quad E'(a) = I_2, \quad E'(b) = J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.
 \end{aligned}$$

All vertex dimensions and all individual arrow ranks agree.

Proposition 11.2 (Scalar incompleteness). *The diagrams E and E' are not isomorphic, although they have identical vertex dimensions and arrow ranks.*

Proof. An isomorphism would be a pair of invertible maps (S_p, S_q) satisfying

$$S_q E(a) = E'(a) S_p, \quad S_q E(b) = E'(b) S_p.$$

The first equation gives $S_q = S_p$. The second would therefore conjugate I_2 to J , impossible because

$$\text{rank}(I_2 - I_2) = 0, \quad \text{rank}(J - I_2) = 1.$$

$\square$

The enriched module $\text{Hom}(G_P, E)$ stores the primitive-idempotent decomposition and the independently labelled actions of the two arrow elements. It therefore recovers both maps and distinguishes E from E' . This example is a minimal witness that rank data sufficient for a linear interval family are not a universal finite-diagram encoding.

11.3. Object and bounded-complex descent on a finite poset. Let $P = \{a, b, c\}$ with $a < c$ and $b < c$. In the upper Alexandrov topology, take

$$U_1 = \{a, c\}, \quad U_2 = \{b, c\}, \quad U_{12} = \{c\}.$$

Define local diagrams

$$\begin{aligned} E_1(a) &= \mathbb{k}, & E_1(c) &= \mathbb{k}^2, & E_1(a < c)(1) &= e_1, \\ E_2(b) &= \mathbb{k}, & E_2(c) &= \mathbb{k}^2, & E_2(b < c)(1) &= e_2, \end{aligned}$$

and identify the fibers over c by the identity.

Proposition 11.3 (Explicit object descent). *The local data descend to the global diagram*

$$E(a) = \mathbb{k}, \quad E(b) = \mathbb{k}, \quad E(c) = \mathbb{k}^2,$$

with structure maps $1 \mapsto e_1$ and $1 \mapsto e_2$. The global object is unique up to unique isomorphism compatible with the datum.

Proof. The only overlap is $\{c\}$, and the two-set cocycle is tautological. Apply [Theorem 9.10](#). The equalizer construction glues the common fiber $\mathbb{k}^2$ and leaves the two incoming maps independently labelled. $\square$

For bounded-complex descent, let $K_1^\bullet$ and $K_2^\bullet$ have on c the two-term complex

$$\mathbb{k}[-1] \xrightarrow{1 \mapsto e_1 + e_2} \mathbb{k}^2[0],$$

while at a or b the degree -1 term is zero, the degree-zero term is $\mathbb{k}$, and the structure map is the corresponding coordinate inclusion. Identify the overlap complexes by the identity.

Proposition 11.4 (Explicit bounded-complex descent). *The local complexes descend to a global bounded complex $K^\bullet$ with*

$$\begin{aligned} K^{-1}(a) &= K^{-1}(b) = 0, & K^{-1}(c) &= \mathbb{k}, \\ K^0(a) &= K^0(b) = \mathbb{k}, & K^0(c) &= \mathbb{k}^2, & d_c(1) &= e_1 + e_2. \end{aligned}$$

Proof. Apply [Theorem 9.12](#) degree-wise. The differential and the equation $d^2 = 0$ are compatible on the overlap and therefore descend. $\square$

The example validates object-level globalization and bounded-complex globalization. It does not support a general claim that any family of local equivalence classes descends: actual overlap maps and a valid cocycle remain mandatory.

11.4. A complete Return through both cores. Let C be the filtered chain complex over $\mathbb{k}$ with

$$C_0 = \mathbb{k}x \oplus \mathbb{k}z, \quad C_1 = \mathbb{k}y \oplus \mathbb{k}w, \quad \partial y = x, \quad \partial w = z,$$

and filtration births

$$\beta(x) = 0, \quad \beta(z) = 1, \quad \beta(y) = 2, \quad \beta(w) = \frac{5}{2}.$$

Work on $W = [0, 3)$ and take $\tau = 2$. The degree-zero barcode is

$$\mathcal{B}_0(C) = \{[0, 2), [1, 5/2)\},$$

and degree-one persistent homology is zero.

Define the source-domain predicate

$$L_{0,W,2}(C) : \quad \text{im}(H_0(F_s C) \longrightarrow H_0(F_t C)) = 0 \quad \text{for all } s \leq t \text{ in } W \text{ with } t - s > 2. \quad (5)$$

Theorem 11.5 (Dual-core predicate Return). *For the filtered complex C , the route*

$$C \longrightarrow \text{Core}_P \longrightarrow \text{Core}_{DG} \longrightarrow \text{enriched generator probe} \longrightarrow L_{0,W,2}(C)$$

passes, and

$$\begin{aligned} L_{0,W,2}(C) &\iff T_2^{\text{bar}} W_W P_0(C) = 0 \iff M = 0 \\ &\iff E_{\Gamma_W} = 0 \iff \text{RHom}(G_{A_{[\ell]}}, E_{\Gamma_W}) \simeq 0. \end{aligned}$$

The Return strength is exactly the source predicate (5).

Proof. Both bars have length at most two, so threshold deletion at $\tau = 2$ removes both. The strict span condition $t - s > 2$ is exactly the source interpretation of the absence of bars longer than two. The breakpoint set $\{0, 1, 2, 5/2\}$ with the right-open window boundary at 3 determines an endpoint-complete atom grid without evaluating the excluded endpoint. The finite-grid equivalence transports the zero persistence object to the zero linear diagram, and the projective generator reflects zero. The source interpretation closes the predicate Return. $\square$

11.5. Return ceiling. The preceding route does not reconstruct the filtered chain complex.

Proposition 11.6 (Sharp ceiling of the predicate route). *The route in Theorem 11.5 returns the long-span degree-zero predicate, the repaired degree-zero persistence zero object, and the corresponding zero finite diagram. It does not return the source filtered chain complex, its full persistent homology, or its filtered derived equivalence class.*

Proof. Let B be a filtered complex with one degree-one generator born at zero, no degree-zero generator, and zero differential. Then

$$P_0(C \oplus B) \cong P_0(C),$$

so the repaired degree-zero profile, finite diagram, generator probe, and predicate Return are unchanged. However,

$$P_1(C) = 0, \quad P_1(C \oplus B) \neq 0.$$

Thus the sources are not filtered quasi-isomorphic, and the degree-zero route cannot be promoted to full source reconstruction. $\square$

This is the role of a typed Return ceiling. The route succeeds completely at predicate strength and fails to determine a stronger target for a precise information-theoretic reason.

11.6. Validation by paired positive and negative cases. The examples above suggest a general validation principle:

A route is validated only when a positive case closes at the declared strength, a negative case fails after removal of a load-bearing hypothesis, and the ceiling is stated as precisely as the positive conclusion.

The principle prevents benchmark backflow. A successful finite example does not enlarge the theorem's source class, weaken its hypotheses, or promote it to an external universal statement.

TABLE 5. Demonstrations, outputs, and prohibited strengthenings.

Case	Certified output	Prohibited strengthening
Rank table on $A_{[3]}$	exact interval multiplicities and object class	arbitrary filtered source or non-linear diagram
Parallel-arrow diagrams	enriched object and morphism reconstruction	completeness of scalar dimensions/ranks
V-shaped Alexandrov cover	global object and bounded complex	descent from pairwise classes without maps/cocycle
Dual-core filtered example	exact long-span degree-zero predicate	source complex or full filtered-derived reconstruction

12. SCOPE, CONTRIBUTION, AND RESEARCH PROGRAM

12.1. What the kernel proves. The mathematical kernel established in this paper is the following source-bound composition.

$$\begin{aligned}
&\text{high-dimensional source } X \xrightarrow{\text{purpose-faithful } \mathfrak{R}_{AK}} \text{structured shadow } F_X, \\
&\text{structured shadow } F_X \xrightarrow{\text{repair/certify}} \text{finite Collapse certificate}, \\
&\text{finite Collapse certificate} \xrightarrow{\text{P2DG}} \text{actual finite diagram}, \\
&\text{actual finite diagram} \xrightarrow{\text{effective descent}} \text{global object} \xrightarrow{\text{typed Return}} E_\sigma(X).
\end{aligned}$$

The first arrow is a source-specific purpose-faithfulness theorem and carries the projection loss profile. Repaired Collapse certification is controlled by sharp threshold geometry, actual detector semantics, and a declared certificate model. P2DG is exact on an endpoint-complete finite-grid category and is objectwise only when the source is merely a barcode. Globalization requires actual cocycle and effectivity data. Return requires all nonlocal reductions, coherence, rigidity at the requested strength, and a final source-domain theorem.

The architecture is constructive in two complementary senses. It gives positive finite certificates and explicit reconstruction formulas, and it gives finite causal descriptions of why a route does not close. The latter include true counterexamples, nonzero obstructions, and missing theorem edges, which are not conflated.

12.2. What is outside the stable scope. The stable persistence scope does not contain wild multiparameter classification, uncontrolled infinite rank, or arbitrary nonconstructible modules. The stable diagrammatic scope does not contain infinite categories, uncontrolled locally finite categories, unbounded complexes, arbitrary constructible sheaves, stacks, hypersheaves, microlocal sheaves, or unrestricted $(\infty, 1)$ -categorical descent. Field coefficients are part of the default core; passage to general rings requires separate hypotheses.

The paper does not assert that a finite category algebra is semisimple or has finite global dimension. It does not identify zero detection with thick generation. It does not construct a cone from a metric comparison. It does not infer a cocycle from pairwise isomorphisms, effectivity from readiness, joint conservativity from effectivity, or a full target object from an invariant without rigidity.

No named arithmetic, PDE, geometric-equivalence, or cryptographic open problem is proved here. Source-specific applications may instantiate the architecture, but their source realization and final Return theorems remain independent mathematical work.

12.3. Where the integrated contribution lies. The classical ingredients are individually familiar. The integrated contribution lies in the disciplined passage between them. Persistence stability alone does not provide categorical structure; categorical reconstruction alone does not provide robust finite observation; descent alone does not provide external semantics; and proof engineering alone does not create mathematical truth. The dual-core design makes these dependencies explicit while preserving composability.

Four features are especially consequential.

Purpose-faithful projection. The theory begins before persistence. A source object becomes Core-readable only through a feature-relative, direction-relative realization theorem. This preserves the original high-dimensional projection concept while preventing a shadow from being mistaken for its source. The loss profile is therefore a mathematical input to every later certificate and Return.

Model-relative minimality. Certificate size is not a property of a proposition in isolation. It depends on the information available before querying, the access model, answer encoding, verifier power, and admissible advice. Exact upper and lower bounds become meaningful only after these are fixed. The private-cell and rank-table theorems illustrate two different lower-bound mechanisms: adversarial indistinguishability and linear-algebraic kernel dimension.

Actual-morphism discipline. The architecture insists on actual maps at every stage that consumes naturality, cones, kernels, cokernels, or descent. This blocks a broad family of invalid promotions from scalar or objectwise data. The enriched generator probe then gives a positive replacement: when actual labelled actions are retained, exact reconstruction is possible.

Target-relative Return. A proof may be complete at predicate, invariant, object, exact, or derived strength. Treating these as target-relative rather than arranging all conclusions in one rhetorical hierarchy makes partial results mathematically usable without overstating them. A valid weaker Return is not a consolation verdict; it is an exact theorem in a different target category.

12.4. Next steps. The most important extensions are not further multiplication of document layers. They are formal and mathematical consolidation:

- (i) a proof-assistant formalization of the minimal persistence, finite-diagram, cone, Morita, and strict-descent kernel;
- (ii) a certified compiler from finite-state transition records to stable-image, derived-limit, and apex certificates;
- (iii) a machine-checkable model of finite homotopy-coherent descent with explicit higher cells and effectivity;
- (iv) generator and Morita reconstruction over declared Noetherian bases for perfect finite-presentation objects;
- (v) a uniform calculus for compatible partial Returns and incomparable strength frontiers;
- (vi) a reference verifier that consumes the typed records used by the theorems rather than document prose;
- (vii) one source-specific external theorem whose complete realization-to-Return route is both mathematically new and independently reviewable.

The present kernel is designed to support those extensions conservatively: a new theorem may widen a domain or strengthen a conclusion, but it must do so through a new typed edge rather than by rereading an existing statement more strongly.

APPENDIX A. CANONICAL SOURCE AND THEOREM MAP

The paper uses local theorem numbers for readability. The following table identifies the corresponding canonical v21 statements. The UIDs are taken from CR-v21; expanded proofs remain in the indicated appendices. A row may combine several canonical statements when the paper theorem is a genuine synthesis rather than a verbatim restatement.

Table 6: Canonical source map for the principal results.

Paper result	Canonical v21 UID	Canonical locator	Expanded proof or validation
Threshold-gap stability	ak-hpdst:21.0.0:main:ch2:theorem-2.46	Main p. 51, Theorem 2.46	Foundation A, Theorem 1.37
Zero separation	ak-hpdst:21.0.0:main:ch2:theorem-2.53	Main p. 54, Theorem 2.53	Foundation A, Theorem 1.44
Repaired-zero transport	ak-hpdst:21.0.0:main:ch2:theorem-2.56	Main p. 54, Theorem 2.56	Foundation A, Theorem 1.47
Finite Pareto frontier	ak-hpdst:21.0.0:main:ch3:theorem-3.86	Main p. 96, Theorem 3.86	Foundation B, Theorem 2.67
Query lower bound	ak-hpdst:21.0.0:main:ch3:theorem-3.93	Main p. 98, Theorem 3.93	Foundation B, Theorems 2.74 and 2.77
Right-unbounded no-go	ak-hpdst:21.0.0:main:ch3:theorem-3.97	Main p. 99, Theorem 3.97	Foundation B, Theorems 2.79–2.80
Finite-state inverse limit	ak-hpdst:21.0.0:main:ch5:theorem-5.84	Main p. 159, Theorem 5.84	Foundation D, Theorem 4.114
Apex agreement	ak-hpdst:21.0.0:main:ch5:theorem-5.85	Main p. 160, Theorem 5.85	Foundation D, Theorem 4.115
Finite-to-infinite soundness	ak-hpdst:21.0.0:main:ch5:theorem-5.88	Main p. 160, Theorem 5.88	Foundation D, Theorem 4.119
Detector-atlas globalization	ak-hpdst:21.0.0:main:ch6:theorem-6.55	Main p. 185, Theorem 6.55	Foundation E, Theorem 5.115
Finite-grid equivalence	ak-hpdst:21.0.0:main:ch4:theorem-4.15	Main p. 112, Theorem 4.15	Foundation C
Möbius reconstruction	ak-hpdst:21.0.0:main:ch4:theorem-4.25	Main p. 115, Theorem 4.25	Foundation C
P2DG handoff	ak-hpdst:21.0.0:main:ch4:theorem-4.98	Main p. 132, Theorem 4.98	Foundation C; DG consumer audit
Generator zero reflection	ak-hpdst:21.0.0:main:ch7:theorem-7.14	Main p. 205, Theorem 7.14	DG, Theorem DG.5.1
Cone criterion	ak-hpdst:21.0.0:main:ch7:theorem-7.16	Main p. 205, Theorem 7.16	DG, Theorem DG.6.1
Finite Morita reconstruction	ak-hpdst:21.0.0:main:ch7:theorem-7.21	Main p. 206, Theorem 7.21	DG, Theorem DG.7.6
Alexandrov realization	ak-hpdst:21.0.0:main:ch7:theorem-7.27	Main p. 207, Theorem 7.27	DG, Theorem DG.9.2
Strict sheaf descent	ak-hpdst:21.0.0:main:ch7:theorem-7.31	Main p. 208, Theorem 7.31	DG, Theorem DG.10.2
Bounded-complex descent	ak-hpdst:21.0.0:main:ch7:theorem-7.34	Main p. 209, Theorem 7.34	DG, Theorem DG.11.2
Global zero certificate	ak-hpdst:21.0.0:main:ch7:theorem-7.42	Main p. 211, Theorem 7.42	DG, Theorem DG.14.2
Global isomorphism certificate	ak-hpdst:21.0.0:main:ch7:theorem-7.43	Main p. 211, Theorem 7.43	DG, Theorem DG.14.3

Paper result	Canonical v21 UID	Canonical locator	Expanded proof or validation
Core-DG package	ak-hpdst:21.0.0:main:ch7:theorem-7.54	Main p. 213, Theorem 7.54	DG, Theorem DG.17.2
Purpose-faithful realization	ak-hpdst:21.0.0:main:ch9:definition-9.90	Main p. 281, Definition 9.90	Main Chapter 9; Problem Interface contracts
Final AK architecture	ak-hpdst:21.0.0:main:ch9:proposition-9.127	Main p. 289, Proposition 9.127	Main Chapter 9; Problem Interface/Companion contracts
Projection-and-return principle	ak-hpdst:21.0.0:main:ch9:theorem-9.128	Main p. 289, Theorem 9.128	Main Chapter 9; CR-v21 registration
Causal blocker antichain	ak-hpdst:21.0.0:ub:UB-v21-DE R-04	Appendix UB p. 23, Theorem 14.4	Appendix-local proved theorem; not a second Main identity
Unified Bridge closure	ak-hpdst:21.0.0:main:ch9:theorem-9.47	Main p. 269, Theorem 9.47	HT, Theorem 11.2; UB components
Dual-core predicate Return	ak-hpdst:21.0.0:main:ch10:theorem-10.27	Main p. 298, Theorem 10.27	Chapter 10 validation record
Exact rank-query complexity	ak-hpdst:21.0.0:main:ch10:theorem-10.30	Main p. 299, Theorem 10.30	Chapter 10 and UB validation

APPENDIX B. COUNTEREXAMPLE SUITE AND PROOF BOUNDARIES

The positive theorems are paired with counterexamples that isolate distinct invalid promotions. The examples are not interchangeable warnings: each blocks a different source-to-target type error.

B.1. Threshold crossing. [Theorem 6.3](#) shows that a small perturbation may move a bar across the hard threshold. It blocks global non-expansiveness and any argument that consumes only one-sided clearance. The repair is the sharp two-sided band in [Theorem 6.4](#).

B.2. Incomplete finite observation. [Theorem 6.13](#) shows that a finite stencil cannot certify a right-unbounded translation family. It blocks the inference from finite observed stability to eventual zero or eventual constancy. The repair is a proved global birth bound, finite birth set, cofinal detector, or finite-to-infinite theorem.

B.3. Omitted relations. [Theorem 9.1](#) shows that graph data and arrow matrices do not determine a category-valued functor unless the relation table is present. It blocks reconstruction from a free-path presentation when the target category has additional relations.

B.4. Scalar probes. [Theorem 11.2](#) shows that vertex dimensions and individual arrow ranks do not reconstruct a diagram with independently labelled parallel arrows. The repair is the enriched module action of the finite generator.

B.5. Objectwise isomorphism. [Theorem 9.7](#) shows that objectwise isomorphic values need not admit a natural isomorphism. It blocks cone formation and morphism Return from object-class data.

B.6. Pairwise overlap maps. [Theorem 9.11](#) shows that pairwise isomorphisms may violate the triple-overlap cocycle. It blocks descent from pairwise matching alone. The repair is an actual descent datum and an effectivity theorem in one selected lane.

B.7. Effectivity without conservativity. An effectivity procedure may construct a global object while the selected restriction functor fails to reflect zero. It blocks global-zero conclusions from effectivity alone. The repair is exact restriction plus joint conservativity, as used in [Theorems 9.14](#) and [9.15](#).

B.8. Predicate Return without source reconstruction. [Theorem 11.6](#) gives two filtered sources with the same degree-zero persistence output and different degree-one persistence. It blocks promotion of a successful predicate Return to full source reconstruction. The repair would require a stronger observation and a source-specific rigidity theorem.

B.9. Finite-prefix and infinite-target separation. A finite prefix of a tower may be extended by inequivalent tails. Therefore a finite observation of stability, vanishing, or periodicity cannot identify an inverse limit or apex without a certified tail theorem. This boundary is formally parallel to [Theorem 6.13](#): both are indistinguishability results in which the omitted region carries the counterexample.

APPENDIX C. COMPACT RECORD SCHEMAS

The mathematical article does not reproduce the complete machine schemas of the v21 corpus. The following compact forms record the fields used by the proofs.

C.1. Certificate record. A theorem-facing finite certificate may be represented by

$$\mathcal{C}_X = (\text{id}, \text{family}, \text{stage}, \Theta, \mathfrak{M}, \text{transcript}, \text{verdict}, \text{cost}, \text{theorems}, \mathcal{A}).$$

Omission of family membership, stage, completeness, verifier semantics, or artifact identity makes theorem-facing consumption undefined.

C.2. P2DG record. A reconstruction-ready P2DG record contains

$$(\text{source}, \text{stage}, W, \tau, \Gamma, \text{category}, \text{relations}, \text{variance}, \\ F_{\text{obj}}, F_{\text{mor}}, \text{exact}, \text{reflect}, \text{info}, \text{loss}, S, \mathcal{A}).$$

The morphism action is independently typed. A barcode-only object branch cannot be consumed by a cone or descent node.

C.3. Descent record. A compact descent input is

$$(\{E_i\}, \{g_{ij}\}, \{c_{ijk}\}, \{h_\sigma\}, \mathcal{W}, \mathcal{C}_{\text{obj}}, \mathfrak{M}_{\text{coh}}, T_{\text{eff}}, \mathcal{A}).$$

Here c_{ijk} denotes literal cocycle identities or specified invertible two-cells, according to the selected model, and h_σ contains every higher cell required by that model.

C.4. Bridge record. A complete target route is represented by

$$(\text{id}, S, \mathcal{T}, \Gamma, S, \text{nodes}, \text{edges}, \mathfrak{D}_{\text{Bridge}}, \text{primary result}, \mathcal{A}).$$

The target identity, assumptions, and strength belong to the mathematical statement. Changing one of them creates a different Return proposition.

DATA AND SOURCE AVAILABILITY

No empirical data set is used. The mathematical source artifacts, canonical locators, and claim identities used by this article are listed in [Appendix A](#) and in the v21 final claim register [\[21\]](#).

VERSIONING NOTE

This manuscript is a standalone scholarly projection of AK-HPDST v21.0.0, re-audited to separate mathematical implication, source-controlled licensing, and Appendix-local proof-DAG results. Its local numbering is for exposition; canonical identities remain those listed in Appendix A. Editorial compression in this article does not enlarge the scope or strength of the source statements.

REFERENCES

- [1] I. Assem, D. Simson, and A. Skowroński, *Elements of the Representation Theory of Associative Algebras, Vol. 1*, London Mathematical Society Student Texts 65, Cambridge University Press, 2006.
- [2] J. A. Barmak, *Algebraic Topology of Finite Topological Spaces and Applications*, Lecture Notes in Mathematics 2032, Springer, 2011.
- [3] U. Bauer and M. Lesnick, Induced matchings and the algebraic stability of persistence barcodes, *J. Comput. Geom.* **6** (2015), no. 2, 162–191.
- [4] M. B. Botnan and M. Lesnick, An introduction to multiparameter persistence, in *Representations of Algebras and Related Structures*, EMS Series of Congress Reports, European Mathematical Society, 2020, 77–150.
- [5] F. Chazal, V. de Silva, M. Glisse, and S. Oudot, *The Structure and Stability of Persistence Modules*, SpringerBriefs in Mathematics, Springer, 2016.
- [6] W. Crawley-Boevey, Decomposition of pointwise finite-dimensional persistence modules, *J. Algebra Appl.* **14** (2015), no. 5, 1550066.
- [7] P. Gabriel, Unzerlegbare Darstellungen I, *Manuscripta Math.* **6** (1972), 71–103.
- [8] S. I. Gelfand and Y. I. Manin, *Methods of Homological Algebra*, 2nd ed., Springer Monographs in Mathematics, Springer, 2003.
- [9] M. Kashiwara and P. Schapira, *Categories and Sheaves*, Grundlehren der mathematischen Wissenschaften 332, Springer, 2006.
- [10] M. Lesnick, The theory of the interleaving distance on multidimensional persistence modules, *Found. Comput. Math.* **15** (2015), 613–650.
- [11] S. Mac Lane, *Categories for the Working Mathematician*, 2nd ed., Graduate Texts in Mathematics 5, Springer, 1998.
- [12] S. Y. Oudot, *Persistence Theory: From Quiver Representations to Data Analysis*, Mathematical Surveys and Monographs 209, American Mathematical Society, 2015.
- [13] C. A. Weibel, *An Introduction to Homological Algebra*, Cambridge Studies in Advanced Mathematics 38, Cambridge University Press, 1994.
- [14] S. Mac Lane and I. Moerdijk, *Sheaves in Geometry and Logic: A First Introduction to Topos Theory*, Universitext, Springer, 1992.
- [15] E. Miller and B. Sturmfels, *Combinatorial Commutative Algebra*, Graduate Texts in Mathematics 227, Springer, 2005.
- [16] A. Kobayashi, *AK High-Dimensional Projection Structural Theory, Version 21.0.0: Finite Globalization, Unified Bridge Calculus, and Certificate Complexity*, Main Chapters 1–10, source-controlled research manuscript, 2026.
- [17] A. Kobayashi, *AK-HPDST v21.0.0, Foundation Core-P and Verification Appendices A–G*, expanded proofs, certificate models, schemas, and regression records, 2026.
- [18] A. Kobayashi, *AK-HPDST v21.0.0, Appendix DG: Finite Diagrams, Projective Generators, Cones, Morita Reconstruction, Alexandrov Realization, and Effective Descent*, 2026.
- [19] A. Kobayashi, *AK-HPDST v21.0.0, Appendix HT: Higher Transfer, Unified Bridge Composition, and Typed Return*, 2026.
- [20] A. Kobayashi, *AK-HPDST v21.0.0, Appendix UB: Universal Bridge Component Library and Validated Return Portfolio*, 2026.
- [21] A. Kobayashi, *AK-HPDST v21.0.0, Appendix CR-v21: Final Claim Register*, 2026.